# LoCoBench: A Benchmark of Coherence Ladders for Evaluating Logical Coherence Measures

**Anonymous authors**
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## Abstract

Measuring whether a long-form response *hangs together* — whether the claims it asserts form a consistent argument rather than a bag of individually checkable facts — requires evaluation data that no existing resource provides. Two documents that differ in coherence almost always differ in topic, length and content too, so a score difference between them cannot be attributed to coherence; human ratings supply a target but not a controlled contrast; and no resource supplies the *relation graph* that would let a measure's scoring errors be separated from its relation-extraction errors. We introduce **LoCoBench**, built from *coherence ladders*: families of five responses over one claim set that differ in one controlled structural respect, whose relative ordering is declared before any measurement, and whose every rung carries its own machine-readable gold relation graph. Four ladder types vary conflict structure, entailment-chain integrity and assertion order, and include a control whose rungs must all score *identically*. Families are generated by a five-stage prompted pipeline and admitted only after an independent cross-vendor committee, which excludes the generator, certifies that the prose realizes its planned relations, reads as expert prose, and asserts every planned claim. The released corpus comprises $100$ families and $500$ responses over $36$ topics, with $4{,}761$ gold edges and gold data for six discourse senses — including exhaustive alternatives and resolved concessions — that no existing dataset labels. Scored from its gold graphs, a probabilistic coherence measure satisfies $89.1\%$ of $1{,}454$ declared constraints, and both invariance ladders are exactly flat. On a $14$-family development set the same measure scored from mined graphs satisfies $34.7$–$37.1\%$ and fails every invariance check, and the best relation extractor reaches $51.9\%$ coupling-level F1: the bottleneck is relation extraction, not scoring. We also report four construction defects that passed every content gate and were found only by scoring the gold graphs. A pilot human study finds that readers agree with the declared ordering on $8/10$ increase comparisons ($\alpha = 0.75$), and identifies one ladder type whose declared invariance readers reject; we specify a human validation study that keeps every annotator under three hours, release all prompts verbatim, and release every generation attempt with its outcome.

## 1 Introduction

Long-form factuality evaluation decomposes a response into atomic claims and verifies each against retrieved evidence (Min et al., 2023; Wei et al., 2024; Song et al., 2024). The paradigm cannot see whether the claims form an argument. A response asserting both that no one was harmed in an incident and that three people died contains one true claim and one false one, and per-claim verification reports exactly that, without registering that the response contradicts itself; a chain of true statements joined by a spurious "therefore" has no unsupported atom in it at all. Xie et al. (2026) quantify the omission: on logic-related samples, SAFE, VeriScore and FactCheck-GPT miss the logical dimension on $60\%$, $78\%$ and $48\%$ of cases respectively.

Measures of *logical coherence* that address this exist or can be built — from entity grids (Barzilay & Lapata, 2008) and their neural successors (Li & Jurafsky, 2017) through discourse-representation scores (Zhao et al., 2023) and reasoning-chain metrics (Golovneva et al., 2023) to LLM judges (Liu

et al., 2023) and probabilistic models over claim-relation graphs. What does not exist is the data to evaluate them on. This paper builds it.

**Why ordinary corpora will not do.** The natural experiment — score a set of documents and correlate against human coherence ratings — has three compounding defects. *Confounding*: two documents that differ in coherence differ in topic, length, register and claim content as well, so a measure that tracks length or claim count can correlate respectably with ratings for reasons unrelated to coherence. *No ground truth for the mechanism*: a rating says a document is incoherent, not *which claims* are in tension, so a measure that is right for the wrong reason is indistinguishable from one that is right. *No separation of error sources*: a measure that builds an explicit relation graph both extracts relations and scores them, and a rating cannot say which of the two failed.

**What LoCoBench provides instead.** We build families of responses — *ladders* — over a single claim set, where the rungs differ only in a controlled structural respect and their relative ordering is *declared before any measurement*. A conflict ladder, for example, holds topic, claims and every unperturbed relation fixed while varying only the conflict structure. Because the rungs share a claim set, satisfying the declared ordering is attributable to coherence and not to content; and because two ladder types declare that every rung must score *identically*, the benchmark can catch a measure that tracks the fact of having been edited rather than coherence. Every rung ships with a gold relation graph — which claims are related, by which discourse sense, in which direction, at what strength, and whether the relation genuinely holds or is a planted error — so a measure can be run once on its own extracted relations and once on the gold graph, and the difference is the extractor's contribution, isolated by construction.

**A documented gap in existing annotation.** Six of the twelve discourse senses in our scheme — Instantiation, Restatement, Contrast, Concession, Alternative and Disjunction — have no gold data in any existing dataset we are aware of. Two of them are what adjudicative prose turns on: an Alternative is a pair of *exhaustive* competing claims of which exactly one holds, and a Concession is a tension the text itself resolves. A measure that conflates mere opposition with exhaustive opposition mis-scores every incident investigation and contested judgement, and it cannot be caught doing so without labelled examples of both. Supplying them is a primary purpose of the corpus.

**Contributions.** We make four contributions, each stated with the evidence offered for it.

1. **Coherence ladders as an evaluation design** (§3–§4). A formal definition of a ladder; a two-level annotation scheme whose discourse senses compile deterministically to truth-functional couplings; four ladder types that vary conflict structure, entailment-chain integrity and assertion order; and three classes of declared ordering constraint, including predicted inversions and exact invariances, which let the benchmark detect a measure that counts edits rather than measuring coherence.
2. **The LoCoBench corpus** (§7–§8). 100 families and 500 responses over 36 topics in four domains, with per-rung gold relation graphs, 1,928 planted invalid relations, declared non-relations as a genuine negative set, and gold data for six discourse senses no existing dataset labels.
3. **A reproducible generation and committee-validation pipeline** (§5–§6, Appendix A). Five prompted generation stages, eleven perturbation calls with declared effects on the gold graph, and three validators run by a cross-vendor committee that excludes the generator. Every acceptance threshold is in one table, every prompt — generation, validation, retry feedback, relation mining and judge baselines — is reproduced verbatim and checked against the code by a test, and every generation attempt is released with its outcome and the gate that rejected it.
4. **Validation of the benchmark, by measurement and by people** (§10–§11). Evidence that the ladders discriminate between relation extractors and between coherence measures; a localization of current measures' failures in relation extraction rather than in scoring; four construction defects that only the gold arm exposed, with their fixes; a pilot human study; and a pre-registered validation study, under three hours per annotator, testing label validity, edit fidelity and perceived ordering.

## 2 RELATED WORK

**Factuality benchmarks and their blind spot.** FActScore (Min et al., 2023) pairs biography prompts with a knowledge source and scores per-atom precision; SAFE (Wei et al., 2024) contributes LongFact-Objects and search-augmented verification; VeriScore (Song et al., 2024) restricts to verifiable claims. Each supplies claims and evidence, and none supplies inter-claim structure, so none can serve as evaluation data for a coherence measure. Xie et al. (2026) is the closest prior art; it constructs data for *logic-related* fact-checking and labels dependencies between claims, but as a classification target rather than as a graded ordering over controlled variants.

**Perturbation-based coherence evaluation.** Evaluating coherence models against synthetic perturbations is long established: entity-grid work (Barzilay & Lapata, 2008) discriminates a document from its sentence permutations, and neural coherence models (Li & Jurafsky, 2017) add insertion and replacement perturbations. LoCoBench inherits the methodology and changes one thing: the perturbation is an operation on a *declared relation graph*, whose effect on the graph is computed and recorded per rung, rather than an operation on text whose effect on coherence is assumed. The benchmark can therefore say not only that rung $k{+}1$ should score above rung $k$ but which relation changed, and — for meaning-preserving edits — that two rungs must score *identically*. Shuffling alone cannot express that distinction; indeed our pilot study (§11) suggests that the classic shuffle test measures something other than logical coherence.

**Discourse annotation as precedent.** Our sense inventory is grounded in PDTB 3.0 (Prasad et al., 2019) and RST (Mann & Thompson, 1988). Both establish that discourse relations can be annotated from surface cues, which is what makes the relation-recovery gate of §6 a reasonable demand. PDTB annotates naturally occurring text, where the distribution of relations is whatever the corpus happens to contain — which is why six of our senses lack gold data. A generated corpus can *require* the rare relations, and ours does.

**LLM-generated benchmarks, LLM judges, and their validation.** Generating evaluation data with a language model is now common, and its known risks are circularity (the generator's biases become ground truth), leakage (artifacts revealing the construction), and unverifiable labels. We address these structurally — a committee that excludes the generator, a leakage validator, and a gate requiring an independent model to *recover* the planned relations from the prose — and empirically, with human studies (§11) whose design follows standard practice for annotation reliability (Krippendorff, 2018) and paired-comparison modelling (Bradley & Terry, 1952). Justifying the committee as a stand-in for human annotators, by the alternative-annotator test (Calderon et al., 2025), is left to future work (§11.2).

## 3 THE ANNOTATION SCHEME

An item's gold annotation is a graph over its claims. Edge labels come from a two-level scheme: an interpretable *discourse sense* that an annotator can read off surface cues, and an inferential *coupling* that says what the relation asserts about the joint truth of its endpoints. A deterministic map connects the two levels, and that map — not the annotator and not the generator — is the authority on what a sense means.

### 3.1 LEVEL ONE: INFERENTIAL COUPLINGS

**Definition 1** (Inferential couplings)**.** *A coupling $\tau$ is specified by the set $V(\tau) \subseteq \{0,1\}^2$ of joint truth assignments $(a_s, a_t)$ of its source and target claims that it* penalizes*:*

| *Coupling* | *Reading* | *Directed* | *Penalized set $V(\tau)$* |
|---|---|---|---|
| ENTAILMENT | $a_s \Rightarrow a_t$ | *yes* | $\{(1,0)\}$ |
| CONTRADICTION | $a_s \Rightarrow \neg a_t$ | *yes* | $\{(1,1)\}$ |
| EQUIVALENCE | $a_s \Leftrightarrow a_t$ | *no* | $\{(0,1),(1,0)\}$ |
| EXCLUSIVE | *exactly one of $a_s, a_t$ holds* | *no* | $\{(0,0),(1,1)\}$ |
| CO_NECESSITY | *at least one of $a_s, a_t$ holds* | *no* | $\{(0,0)\}$ |
| NONE | *no truth coupling* | — | $\varnothing$ |

Table 1: The compile map $\mathcal{C}$ of Def. 2, with the number of gold edges of each sense over all rungs of the released corpus. *Ordering-only* senses record assertion order but carry **no truth coupling**, so they produce no factor in any scorer that consumes the scheme. **Rev.** marks the senses whose edge can be reversed without changing any factor the scorer builds (§5.6); this is a property of the factor table, not of the *dir.* flag, so the symmetric sense Contrast does not qualify. **Bold** marks the six senses for which we are aware of no gold data in any existing dataset.

| Level-2 sense | $\mathcal{C}$ coupling | prior | dir. | flag | rev. | gloss | edges |
|---|---|---|---|---|---|---|---|
| Cause-Effect | ENTAILMENT | $\perp$ | yes | | | source brings target about | 667 |
| Effect-Cause | ENTAILMENT | $\perp$ | yes | | | source is effect, target cause | 51 |
| Evidence | ENTAILMENT | $\perp$ | yes | | | source is evidence for target | 660 |
| Condition | ENTAILMENT | $\perp$ | yes | | | source conditions target | 36 |
| **Instantiation** | ENTAILMENT | $\perp$ | yes | | | target instantiates source | 70 |
| **Restatement** | EQUIVALENCE | 0.90 | no | | ✓ | definitional identity | 560 |
| **Contrast** | CONTRADICTION | $\perp$ | no | | | opposed, *not* exhaustive | 716 |
| **Concession** | CONTRADICTION | $\perp$ | yes | conc. | | a tension the text resolves | 500 |
| **Alternative** | EXCLUSIVE | $\perp$ | no | | ✓ | exhaustive: exactly one holds | 491 |
| **Disjunction** | CO_NECESSITY | $\perp$ | no | | ✓ | at least one holds | 500 |
| Precedence | NONE | — | yes | ord. | ✓ | temporal order only | 459 |
| Succession | NONE | — | yes | ord. | ✓ | temporal order only | 51 |
| Total | | | | | | | 4,761 |

*The two couplings whose penalized set contains the both-true world,* $\{$CONTRADICTION, EXCLUSIVE$\}$*, are the* conflict *couplings.*

The distinction between CONTRADICTION and EXCLUSIVE is the one the scheme exists to draw. CONTRADICTION forbids only the both-true world: two claims that cannot both hold but might both fail. EXCLUSIVE additionally forbids the both-false world, which is what an exhaustive pair asserts. CO_NECESSITY is the dual, forbidding only the both-false world, which is what a disjunction asserts. A scheme offering only CONTRADICTION can express neither, and a measure evaluated against such a scheme is untested on precisely the relations adjudicated disputes turn on.

### 3.2 Level two: discourse senses, and the compile map

**Definition 2** (Compile map)**.** *The compile map $\mathcal{C}$ assigns to each discourse sense a coupling, a strength prior ($\perp$ meaning "use the estimated value"), and the flags* directed*,* concession *and* ordering-only*. It is total and deterministic, and is given in Table 1.*

Five senses collapse onto ENTAILMENT, a deliberate loss of information at the inferential level: whether one claim causes another or merely evidences it makes no difference to what the pair asserts about joint truth. The sense is retained alongside the coupling so that error analysis can distinguish "related, but mislabelled" from "not related", and so that the corpus serves work on the discourse layer for its own sake.

**The compile map is enforced.** A gold relation carries both a sense and a coupling, which invites inconsistency. The schema validator rejects any item in which coupling $\neq \mathcal{C}(\text{sense})$, and re-derives directedness, ordering-only status and exhaustiveness from the sense rather than trusting the generator's assertion of them. This matters for a generated corpus: an LLM asked for both a sense and a coupling will occasionally emit a pair no reading of the scheme licenses, and keeping such an edge would place a factor in the graph that its sense does not justify. Every gold edge in the corpus therefore uses one of the twelve $(\text{sense}, \text{coupling})$ combinations of Table 1.

## 4 Coherence ladders

**Definition 3** (Coherence ladder)**.** *A* ladder *is a tuple $(\mathcal{P}, \langle \rho_0, \ldots, \rho_4 \rangle, \Gamma)$ where $\mathcal{P}$ is a base* relation plan *— a selected claim set with a declared relation graph over it — each rung $\rho_k$ is a sequence of perturbation operator calls applied to $\mathcal{P}$, yielding a response text together with its* own *gold relation*

Table 2: The four ladder types. Each holds the claim set, the topic and every unperturbed relation fixed, varying one structural dimension. $n$ is the number of families in the released corpus, the generator's proportional allocation at 100 families. The ORDER and CONTROL patterns are declared with respect to *logical* coherence: their edits change no truth coupling, so the gold graph, and any score computed from it, must not move. Whether *readers* agree is an empirical question that the pilot study answers differently for the two types (§11).

| Type | Dimension varied | Rungs, $\rho_0 \to \rho_4$ | Declared pattern | $n$ |
|---|---|---|---|---|
| CONFLICT | conflict structure | `worse`, `base`, `concession_resolved`, `fix_one_conflict`, `coherent` | increasing, with one predicted dip | 46 |
| CHAIN | entailment-chain integrity | `break_3`, `break_2`, `break_1`, `base`, `coherent` | increasing; $\log Z$ decreasing until `base` | 21 |
| ORDER | assertion order | `shuffle_full`, `shuffle_block`, `shuffle_adjacent`, `base`, `ordering_only` | **flat throughout** | 21 |
| CONTROL | nothing (meaning-preserving) | `ordering_only`, `direction_reversal`, `base`, `direction_reversal`, `ordering_only` | **flat throughout** | 12 |

*graph, and* $\Gamma$ *is a set of constraints on the relative ordering of the rungs' scores. A ladder is* admitted *only as a whole (§6).*

**Whole-family admission.** A ladder is admitted entire or not at all. Four of the five rungs carry no standalone claim, only a claim relative to their neighbours, so a ladder missing a rung is not a partial success but a broken comparison. Every gate therefore rejects at family granularity.

Table 2 gives the four types. CONFLICT is the reference ladder: only the conflict structure varies. A spurious relation makes the lowest rung worse than the base; the upper rungs resolve the conceded tension and then drop one and two conflict edges. CHAIN breaks $N \in \{3, 2, 1\}$ links of a chain of *valid* entailment relations that its plan is required to contain, so the defect is a *missing* relation rather than a present conflict — a distinction that matters because a measure built on contradiction detection alone will do well on the first ladder and fail the second. Its top rung drops one planted-invalid conflict.

ORDER and CONTROL are *invariance* ladders. ORDER permutes sentences at three granularities and then exchanges a Precedence for a Succession; CONTROL exchanges Precedence for Succession on its outer rungs and, on its inner rungs, reverses a *coupled* relation whose factor is symmetric in its endpoints (Restatement, Alternative or Disjunction; Table 1). None of these edits changes a factor, so the declared pattern is a flat line and a measure that trends across either has demonstrated that it responds to *having been edited*. The two types are kept separate on purpose. CONTROL edits are meaning-preserving for any reader. ORDER edits preserve every claim and every coupling but can strand an anaphor or put evidence after the conclusion it supports — they preserve *logical* coherence while degrading *discourse* coherence. Keeping the types apart lets the benchmark, and the human study, ask whether a measure (or a reader) can tell those two properties apart.

### 4.1 ORDERING CONSTRAINTS

A family declares its constraints in three classes, each asserted per *readout* of the measure under test, so that a family can declare that one readout must increase between two rungs while another is unconstrained there.

**C1, strict increase.** Consecutive rungs must be strictly ordered.

**C2, predicted inversion or invariance.** A score must *decrease*, or must be *exactly invariant*. This is what distinguishes a declared-ordering benchmark from a monotonicity test. Some readouts are expected to dip when a conceded tension is resolved, because softening a conflict raises the

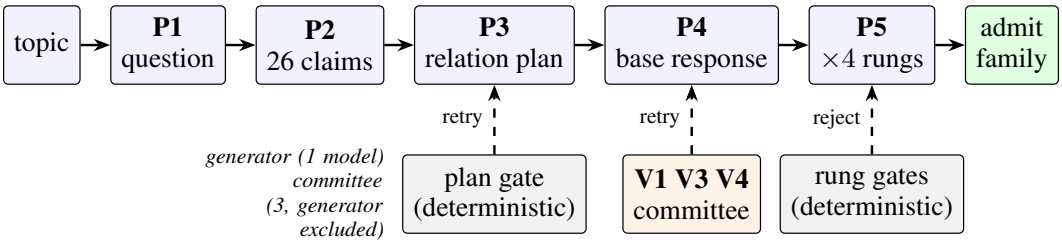



Figure 1: The pipeline. Five generation stages produce a family of five rungs from one topic. The plan is checked by a deterministic gate, the base response by the three-model committee, and each perturbed rung by deterministic gates; gate failures are fed back to the generating stage as a retry, and a family that exhausts its retries is rejected whole and stored with its reason.

probability of the very configuration the conflict forbids; asserting the dip turns a known peculiarity into a falsifiable prediction. The same applies along CHAIN: restoring an entailment link adds a factor that concentrates the joint's mass, so the normalized log partition is predicted to *fall* at each breaking step while the belief readouts rise. This declaration was revised on measurement, and §10 reports the evidence. And a coupling-free edit *must* be exactly flat: C2 then asserts an equality, and a measure that moves at all has been caught reading surface form.

**C3, endpoint separation.** The extreme rungs must separate (or, on the invariance ladders, must not), for all readouts. This is the robust headline claim, insensitive to the fine structure of intermediate rungs.

For the three-readout measure of §9, a CONFLICT family declares 14 constraints (9 C1, 2 C2, 3 C3) and every other family 15: CHAIN has 9 C1, 3 C2 and 3 C3, and ORDER and CONTROL have 12 C2 invariances and 3 C3. The released corpus declares 1,454, and a manifest's constraints are re-derived from the code and checked against it on release. A single-score measure inherits only the readout-independent content: five increase assertions per CONFLICT or CHAIN family (four C1 pairs and the C3 endpoints) and five invariance assertions per ORDER or CONTROL family. A measure is scored on the fraction of declared constraints it satisfies.

## 5 THE GENERATION PIPELINE

Generation is a five-stage prompted pipeline followed by validation, and the stages are separated so that a defect can be attributed to one of them and repaired there. Each stage's prompt is fixed, versioned, and reproduced verbatim in Appendix A; the appendix files are written from the source by a script, and a test fails if any of them differs from the prompt the code issues. Algorithm 1 gives the procedure end to end and Figure 1 its shape.

### 5.1 P1: TOPIC TO QUESTION

Given a topic from the inventory of §7, emit one open-ended question. Three of the nine instructions do real work. The question must *elicit* diverse logical structure, and the prompt enumerates the target relations so that the question is chosen with the downstream plan in mind. It must prefer adjudicative framings — incident investigations, litigation, diagnostic reasoning, attribution disputes, competing scientific hypotheses — because these are where exhaustive alternatives and resolving holdings arise naturally. And it must *not* contain the words "cause", "condition", "alternative" or "contrast", so that the answer's relations are not primed lexically by the question.

### 5.2 P2: QUESTION TO TAGGED CLAIMS

Emit exactly 26 atomic claims, each a single declarative sentence with no connective joining it to another — relations are added at the next stage, so that the claim set and the relation graph are independently controlled. The composition is specified: 14 factually correct claims; 4 plausible-but-false claims; an *exhaustive alternative* pair, of which exactly one must be true; a *disjunctive* pair, of which at least one must be true; an *equivalent* pair; and two *holding* claims in which an authority

**Algorithm 1** Generating one family of type $T$ on topic $t$. $\textsc{Ask}(P, \cdot, g)$ calls prompt $P$, parses the output, applies check $g$, and on a parse or check failure re-issues the prompt with the failure reason appended (Appendix A.9), up to 3 attempts at temperatures $(\text{default}, 0.3, 0.7)$.

$q \leftarrow \textsc{Ask}(\text{P1}, t)$ ▷ open question
$C \leftarrow \textsc{Ask}(\text{P2}, q)$ ▷ 26 tagged claims
$\mathcal{P} \leftarrow \textsc{Ask}(\text{P3}, q, C, T,\ g{=}\textsc{PlanGate}_T)$ ▷ claims, 10 relations, non-relations
$x_{\text{base}} \leftarrow \textsc{Ask}(\text{P4}, q, \mathcal{P},\ g{=}\textsc{Committee})$ ▷ V1, V3, V4
$E_{\text{base}} \leftarrow$ gold graph of $\mathcal{P}$, fields re-derived through $\mathcal{C}$
assign each operator call of each rung of $T$ a distinct eligible target edge; **reject** if too few
**for** each non-base rung $\rho_k$ of $T$, in ladder order **do**
  **for** each call $o$ of $\rho_k$ **do**
    $x \leftarrow \textsc{Ask}(\text{P5}, x, \mathcal{P}, o)$; **reject** unless $x$ changed and drift $\leq 15\%$
  $E_k \leftarrow$ apply the declared edge effects of $\rho_k$'s calls to $E_{\text{base}}$
**reject** if two adjacent rungs have equal edge sets and the upper one claims a change
**reject** if any two rungs have identical text but different edge sets
**reject** unless every item passes the schema validator; otherwise **admit** all five rungs

settles a dispute. Each claim carries a tag recording its role. The disjunctive pair must be phrased as states of the world rather than as findings — a pair of *reports* is trivially compatible and yields no real disjunction — and the false claims must not be signalled by hedging.

### 5.3 P3: CLAIMS TO RELATION PLAN

The planning stage, and the most constrained. Select 14–16 of the claims, renumber them to consecutive assertion positions, and plan exactly ten relations over distinct claim pairs, each with a sense from Table 1, an intended strength band, and a *validity* label. Five requirements shape the output; a deterministic plan gate checks each, and a violation is fed back to the planner as a retry.

*At least two false claims must be selected*, so the response asserts false content alongside true. Without this quota the planner selected none, which would make coherence and factuality perfectly correlated in the data. Two sits just below the expected $\approx 2.3$ under indifferent sampling of 15 from 26 claims with 4 false.

*Five senses are required* — at least one Alternative, Disjunction, Restatement, Concession and temporal ordering — which supplies the scarce senses.

*A ladder-specific quota* then reserves the edges the ladder's own edits will consume, and it is the one part of P3 that differs between ladders, substituted through a placeholder. For CONFLICT, ORDER and CONTROL, *at least four of the ten relations must carry a conflicting sense* (Alternative, Concession or Contrast), at most one of them a resolved Concession, so that the CONFLICT ladder has enough unresolved conflict edges to drop two and resolve a third (§5.6). For CHAIN, *at least three of the six valid relations must be entailments forming a chain*, each link's target the next link's source, which are the links the ladder breaks. Both quotas cannot be imposed on one plan: with ten relations, six of them valid, and five required senses, three valid entailments leave no room for four conflicts. An earlier version imposed the conflict quota on every plan and left CHAIN with no guaranteed chain, which is the defect reported in §10.5.

*Exactly six relations valid and four invalid*, each invalid relation carrying an error kind — wrong_sense, wrong_direction, false_endpoint or spurious. Planted invalid relations are asserted in the prose in the same register as valid ones, and make the corpus a test of *precision*: an extractor that recovers every gold edge including the planted ones has not exercised judgement. The count is stated as a count because a planner asked for "about 60%" rounds unpredictably.

*Four to six declared non-relations*: pairs of selected claims within four positions between which the response must draw no dependence. Precision against all unlabelled pairs is close to meaningless, since most are simply unannotated; a violated *declared* non-relation is a genuine false positive against a real negative set.

Table 3: The six perturbation operators and the eleven parameterized calls realizing them. Each call has a declared effect on the rung's gold relation graph and a declared set of eligible targets (§5.6). The two O6 calls, and direction_reversal on a reversal-invariant relation, change no factor: they are what the ORDER and CONTROL ladders exist to check.

| Op | Calls | Effect on the text |
|---|---|---|
| O1 relabel | wrong_sense, direction_reversal, exhaustiveness_flip | rewrite the connective so the text draws a different sense, reverses the relation, or flips Contrast↔Alternative |
| O2 add conflict | inject_contradiction, spurious_relation | add one sentence directly contradicting a named claim at least two sentences later; or assert a link between two claims the response kept separate |
| O3 break chain | break_chain | remove or negate the link realizing a valid entailment so the chain no longer connects, keeping both endpoints |
| O4 drop relation | drop_relation | remove the connective realizing an edge, leaving the two claims as adjacent assertions |
| O5 resolve | add_resolution, remove_resolution | add or delete the holding sentence that settles a conceded tension |
| O6 reorder | shuffle_order, ordering_only | permute sentences at full, block or adjacent granularity; or exchange Precedence↔Succession preserving chronology — *no coupling changes* |

*Relations are local by default*, preferring pairs within four positions, with long-range pairs permitted when the two claims share a named entity — a deliberate bias toward graphs a reader can recover, so that a V1 failure indicates a defective realization rather than an unreasonable demand.

### 5.4 P4: PLAN TO RESPONSE

Write 550–650 words of prose (floor 500) realizing the plan exactly: assert every planned claim once, in position order, and realize every planned relation so that a careful reader can recover it from the text alone, while reading as publishable expert prose rather than as an annotation walked sentence by sentence. Several instructions resolve that tension. Back-reference uses a pronoun or a noun phrase named after *the thing the claim is about*, not after its status as a finding. Quantifiers, numbers, dates, polarity and modals are *content*, so a claim may be reworded but not weakened. Exhaustive and disjunctive pairs must make their logical strength *explicit*, since a bare "X, or Y" leaves a reader unable to tell Alternative from Disjunction. Temporal orderings must be stated within one sentence. Non-relations are separated by distance and paragraphing rather than by an explicit marker, which would both leak the plan and itself draw a relation. The prompt forbids mentioning the plan, its labels, or the fact that some content is incorrect.

### 5.5 P5: RESPONSE TO RUNG

Apply exactly one perturbation to the current text, changing the minimum number of words and preserving everything else verbatim; a rung composed of several calls applies them in sequence. Table 3 gives the eleven calls, grouped by the six operators they realize. Length drift is capped at 15%, except for the four calls that add or remove a sentence by definition, which get one sentence's further allowance.

### 5.6 DERIVING EACH RUNG'S GOLD RELATION GRAPH

A rung's gold graph is the base plan's graph with that rung's operator calls applied. Four mechanisms make this well defined. *Target assignment*: each call in a rung is assigned a distinct eligible edge, so a rung composing two drop_relation calls drops two *different* edges, and a plan with too few eligible edges is rejected rather than allowed to emit a rung that duplicates its neighbour. Eligibility is call-dependent, and two of its rules were found by measurement (§10.5). break_chain targets only *valid* entailment edges, since breaking a planted-invalid one removes an error and makes the rung more coherent than the base. direction_reversal targets only senses whose factor table is symmetric

in its endpoints. This set is *derived* from the scorer's tables, not listed by hand, so it cannot drift from them, and a test reverses one edge of each sense through the scorer and checks that the score moves exactly when the sense is ineligible. It excludes Contrast: the sense is symmetric, but its with-priors CONTRADICTION table carries the source's prior on one row. Coupled senses are preferred over ordering-only ones, so that CONTROL's two edits never land on the same edge. *Derived-field re-derivation*: when a call changes an edge's sense, its coupling, directedness, ordering-only status and exhaustiveness are re-derived through $\mathcal{C}$. *An adjacency gate*: a family is rejected when two *adjacent* rungs have the same edge set while the upper rung's calls claim a change. Adjacency rather than parentage is the right comparison: every CONFLICT rung declares the base as its parent, so a parent-only check passes a rung that duplicates the rung immediately below it (§10.5). *A text–label gate*: over *all* rung pairs, two rungs with byte-identical prose must carry identical gold edge sets, or at least one rung's labels describe an edit its text does not contain.

### 5.7 THE FAILURE POLICY

What happens when a stage's output is defective is a specification decision. A **retry** re-issues the prompt with the failure reason appended, using one of two fixed notes (Appendix A.9): a *format* note for a parse failure, and a *content* note for a gate failure that tells the model its format was correct and its content must change. The distinction is measured: a single note telling the model to fix its format, sent for a content failure, never repaired the content. Retries are issued at rising temperature, because a byte-identical prompt re-sent at temperature zero cannot produce a different output. Plan-gate and committee failures are retried in this way; parse failures at every stage likewise. A **reject** applies when a family cannot carry its ranking claim — retries exhausted, too few eligible targets, a rung that did not change or drifted too far, or a degenerate adjacent pair — and stores the family with the failing gate, its threshold, the observed value and the raw output. No stage patches a defective artifact into an acceptable one. A further guard concerns output length: P4's target exceeds the default completion-token grant of some gateways, and a truncated response fails every parser for the wrong stated reason, so the output ceiling is raised explicitly.

## 6 COMMITTEE VALIDATION

An item is admitted only if an independent committee certifies it. Three properties of the arrangement matter more than the individual gates.

**The committee excludes the generator and spans vendors.** Validation is run by a panel of models, and any panel member sharing the generating model's identifier is removed before voting. The configuration is refused at load time unless at least three distinct voters remain after exclusion and the panel spans at least three model families, so that agreement cannot be an artifact of shared training. The released corpus was generated by `claude-opus-5-5` and validated by `gpt-5.6`, `gemini-3.1-pro` and `claude-sonnet-5-5`. The generator and one committee member share a vendor; this is one of the dependencies a test of committee validity would measure; §11.2 explains why the validation study defers it.

**Each gate has its own decision rule.** The three validators measure different things and are combined differently (Table 4). **V1** is *any-of*: relation recovery is a claim that a careful reader *can* recover the plan, so one rater clearing both thresholds is sufficient evidence; every rater's rates are recorded alongside the reported one. **V3** is *majority per facet*: a prose defect counts only if a majority of raters report it, so a single unusually strict auditor cannot reject a sound response. **V4** is *majority per claim*: a claim counts as lost only if a majority calls it lost, and a planted-false claim counts as lost if a majority calls it altered.

**The committee validates the base response; rungs are validated by construction.** The committee runs on the base response of each family. The four perturbed rungs are checked by deterministic gates — the text changed, length drift is within bound, the edge effect is non-degenerate between adjacent rungs, identical texts carry identical labels, and the item satisfies the schema — and their gold graphs are derived from the base graph by the declared edge effects rather than re-certified from their prose. Whether a P5 edit realizes its declared effect and nothing more is therefore an assumption of the current pipeline, and the validation study tests it (§11.2, Claim B).

Table 4: Every acceptance threshold, defined in one table in the source and nowhere else. V1 rates are computed over the *valid* planned relations only. The rightmost column is the committee decision rule.

| Gate | Quantity | Threshold | Rule |
|---|---|---|---|
| V1 | valid planned relations recovered, correct coupling | $\geq 0.80$ | any-of |
| V1 | … and with correct sense | $\geq 0.70$ | any-of |
| V3 | fluency, formality, organization (each, of 5) | $\geq 4$ | majority/facet |
| V3 | leakage and hedging span lists | empty | majority/facet |
| V4 | claim coverage over gating statuses | $= 1.00$ | majority/claim |
| V4 | gating statuses | missing, merged | — |
| V4 | … and on a planted-false claim | +altered | majority/claim |
| P3 | claims selected | 14–16 | — |
| P3 | relations planned (prompt asks for 10) | 8–12 | — |
| P3 | declared non-relations | 4–6 | — |
| P3 | fraction valid (prompt asks for $6/10$) | $0.55 \pm 0.15$ | — |
| P3 | false claims selected | $\geq 2$ | — |
| P3 | relation locality (recorded, not enforced) | $\leq 4$ positions | — |
| P5 | length drift | $\leq 15\%$ | — |
| Panel | voters after generator exclusion | $\geq 3$ | — |
| Panel | distinct model families | $\geq 3$ | — |

### 6.1 V1: RELATION RECOVERY — THE LOAD-BEARING GATE

Given the response and its claims — but *not* the plan — a committee model recovers the relations the response itself draws, emitting for each pair a sense, a coupling, a strength band and a resolved flag. Admission requires that at least $0.80$ of the *valid* planned relations come back with the correct coupling and at least $0.70$ with the correct sense. This is the gate that gives the labels meaning: the rest of the pipeline produces a plan, and V1 certifies that the *prose realizes it*. Without it a relation the planner intended and the writer failed to express would remain in the gold graph, and an extractor would be penalized for correctly not finding it.

*The rates are computed over valid relations only.* Planted-invalid relations are deliberately wrong, so requiring them to be recovered would reward exactly the over-assertion the corpus is built to detect, and made the gate unsatisfiable as first specified (the ceiling was $6/10$ against $0.80$). With six valid relations both thresholds require five of six: one relation decides admission, and a coupling threshold above $0.834$ would silently demand unanimity. An admitted item may therefore carry at most one valid gold relation that the reporting rater did not recover; the validation study's claim-pair check (§11.2) measures how often a human reader also fails to see it.

*Recovery is graded at three levels.* A recovered relation is *identified* by its unordered claim pair, independently of its sense; direction, coupling and sense are graded separately. Keying identity on the sense instead would place a planned Concession$(7, 12)$ and a recovered Contrast$(7, 12)$ — same endpoints, both CONTRADICTION — in different key spaces, scoring "recovered with the wrong sense" as "not recovered", which is precisely the distinction V1 exists to measure.

### 6.2 V3: PROSE AUDIT

The response is scored for fluency, formality and organization, each on $1$–$5$ with a floor of $4$, and three span lists are collected: *leakage*, *hedging* and *artifacts*. Leakage and hedging must be empty; artifacts are recorded but do not gate. Leakage is defined narrowly: naming a relation type or sense, naming the plan or its labels, referring to "claims" or "atoms" as the objects the text was assembled from, or any bracketed marker or index. It explicitly does *not* cover the connectives that *realize* relations — "either X or Y", "at least one of these must hold" — because P4 is required to use them; an earlier, broader reading rejected correctly written responses for containing the very markers the pipeline demands. Hedging is a closed list of seven words. Every reported span must be quoted verbatim from the response, which makes a hallucinated objection checkable and discardable.

### 6.3 V4: CLAIM COVERAGE

For each claim, a committee model decides whether the response asserts it, returning asserted, altered, missing or merged with a quoted span. Coverage must be perfect over the *gating* statuses, missing and merged. altered is deliberately non-gating: it is the judgement most sensitive to the rewording P4 licenses, and three of five early V4 rejections were a single claim called altered where the rewording was permitted. The exception is a planted-*false* claim, on which altered also gates (by majority): there the characteristic alteration is the generator correcting the claim to the truth, which leaves a `factual: false` label on prose asserting the opposite (§10.5). The prompt states that two claims joined by a connective are *not* merged, since "either X or Y" asserts both; without that clarification every realized Alternative counted as a merge.

### 6.4 WHAT THE COMMITTEE DOES AND DOES NOT ESTABLISH

**Admission.** On the first full pass after the P5 and committee fixes of §10.5, 88 of the 100 families were admitted at their first attempt. The later fixes did not leave all of them as admitted: 11 CONTROL families kept their base response but had their rungs rebuilt after the reversal fixes, and 17 CHAIN families received a new base response when every CHAIN plan was redone for the entailment-chain requirement. So 71 released families carry the base response admitted on that pass, 60 of them entirely unchanged. Every family was eventually admitted. 7 released families come from a fresh regeneration from P1: 3 after repeated prose-audit rejections, and the 4 withdrawn for corrected false claims (§10.5). Over all 251 logged family attempts, including those superseded by a fix, the rejecting gate was the prose audit in 44 cases (43 on scores, 1 on a leakage span all three raters quoted), an unchanged or over-long P5 rung in 20, and relation recovery in 3. The release includes every attempt and its outcome.

**How decisive the decision rules are.** Each admitted family records every rater's verdict. On relation recovery, each rater alone would have admitted 89–91 of the 100 families, and all three independently clear both thresholds on 77. So the any-of rule is decisive on 23 families and redundant on the rest. On the prose audit the raters are not interchangeable: `claude-sonnet-5-5` scored some facet below the floor on 36 admitted families, the other two raters on 3 each, so the majority rule is what keeps one strict rater from deciding admission. An earlier committee that included `gpt-6-sol` as its OpenAI member, which scored below the floor on 11 of its first 45 votes, combined with the same strict rater into rejecting majorities on 23 families. We replaced it, rather than lowering the floor, so that the gate's threshold stays the one the paper states.

A committee whose members can share biases with the generator and with each other can certify an item that a person would reject, and an admission rate cannot reveal that. The committee's gates are therefore themselves an object of validation: testing each gate's agreement with human judgement needs responses the gates rejected, which the release does not keep, so the validation study defers it (§11.2).

## 7 CORPUS COMPOSITION

Table 5 gives the inventory. Topics and ladder types are assigned to family slots deterministically, so a resumed run assigns the same topic and type to the same slot. Ladder types are allocated in proportion to Table 2. Topics are cycled through the inventory, so that at 100 families every one of the 36 topics contributes at least two families and 28 contribute three; the generator also implements a floor-then-surplus policy (three families per topic, surplus to the five adjudicated topics) that applies from 108 families upward.

**The released corpus.** The release comprises 100 families and 500 responses generated by `claude-opus-5-5`: 46 CONFLICT, 21 CHAIN, 21 ORDER and 12 CONTROL, over all 36 topics in all four domains (31 natural sciences, 29 humanities, 22 engineering and technology, 18 social sciences). Each base plan selects 15–16 claims, 2–3 of them false, and plans 10 relations (6 valid, 4 invalid) and 5 declared non-relations; responses run 567–731 words, mean 643. Across all rungs there are 4,761 gold edges, 4,251 of them edge-producing; 2,833 are valid and 1,928 planted-invalid (768 false_endpoint, 460 spurious, 365 wrong_sense, 335 wrong_direction). Every one of the

Table 5: The topic inventory: 36 topics across four domains. The five **bold** topics are the adjudicated subjects where exhaustive alternatives and resolving holdings occur naturally, and receive the surplus families at corpus sizes above the per-topic floor.

| Domain | Topics | $n$ |
|---|---|---|
| Natural Sciences | Astronomy, Biology, Botany, Chemistry, Geology, Human Biology, **Medicine**, Meteorology, Oceanography, Physics, Zoology | 11 |
| Humanities | Archaeology, Art, Culture, **Ethics**, Gender Studies, History, Linguistics, Literature, Philosophy, Religion | 10 |
| Engineering & Technology | Artificial Intelligence, **Civil Engineering**, Computer Science, Cybersecurity, Electrical Engineering, Mechanical Engineering, Renewable Energy Technologies, Robotics | 8 |
| Social Sciences | Anthropology, Economics, Education, **Law**, **Political Science**, Psychology, Sociology | 7 |

six senses with no prior gold data is represented by at least 70 edges (Table 1). Every family carries five distinct response texts except the CONTROL families, whose ladder is symmetric by design — rungs 0 and 4 apply the same edit, as do rungs 1 and 3 — so each pair carries identical labels and usually identical text: 11 of the 12 have three distinct texts, and one has five, its paired rungs differing only in wording.

**A development set.** An earlier 14-family corpus generated by `claude-opus-5`, with a committee of `gpt-5.6`, `gemini-3.1-pro` and `claude-sonnet-5`, is released separately as a development set. It is the corpus of the pilot human study and of the extraction and baseline results in §10. It predates two of the fixes of §10.5. Its P5 never defined add_resolution, though its generator inferred the operator correctly. More consequentially, its two CHAIN families do not break only entailment links, so its CHAIN results should not be read as testing entailment-chain integrity.

## 8 ITEM SCHEMA AND RELEASE FORMAT

Table 6 gives the schema. The release is four artifacts per corpus: `items.jsonl`, one item per line; `families.json`, the manifest carrying each family's declared ordering constraints; `state.json`, the generation log, including each admitted family's committee verdict; and a `rejected/` directory holding every family that failed a gate with the gate, threshold, observed value and raw output. The rejected set is part of the release because admission rates are auditable only if the denominator is available. Several fields reveal the declared answer (`rung_name`, `expected.perturbation`, the item-id suffix, `factual`); any human or model evaluation must be run on redacted copies, and the release includes the redaction script used in §11.

## 9 THE EVALUATION HARNESS

**Arms: holding the scorer fixed.** An *arm* determines where the relation graph comes from while holding the claims, priors and scoring procedure identical. GOLD takes each item's labelled relations; GOLD_VALID takes only those labelled valid, which quantifies what the planted errors cost; a MINED arm, written `mined:`⟨`model`⟩`:`⟨`policy`⟩, runs a relation extractor on the response text. Because the scorer is identical across arms, any shortfall of a mined arm below the gold arm is attributable to extraction. The gold arm needs no LLM, so it is deterministic and serves as a regression test on the corpus itself.

**What a scorer must expose.** The harness requires only that a measure map a response (and, for graph-based measures, a relation graph over its claims) to one or more scalar readouts. To validate the corpus we use a Markov random field over claim truth variables: one binary variable per claim with unary factor $[1 - \pi_i, \pi_i]$, and one pairwise factor per relation penalizing the assignments $V(\tau)$ of Def. 1. Three readouts are graded: the mean posterior marginal; a consistency probability combining the probability that no conflict is jointly active with the mass of upheld support relations; and the partition function normalized between a conflict-free ceiling and the base network's most probable

Table 6: The item schema. Every field is machine-readable and validated on load; fields marked *derived* are re-computed from the sense through $\mathcal{C}$ rather than trusted as written.

| Field | Type | Meaning |
|---|---|---|
| *Item level* | | |
| `id`, `name` | string | stable identifier and human-readable label |
| `source` | string | generating model |
| `response` | string | the prose |
| `num_atoms` | int | claim count |
| `notes` | string | free-text provenance note |
| *Per claim* (`atoms`) | | |
| `id`, `text` | string | claim identifier and its exact text |
| `label` | string | P2 tag: correct, incorrect, alt-pair, disj-pair, equiv-pair, holding |
| `factual` | bool | whether the claim is true — *false* for planted-false claims |
| `role` | string | structural role in the plan |
| `pos` | int | 1-based assertion position |
| *Per relation* (`relations`) | | |
| `id` | string | edge identifier, referenced by perturbation targets |
| `source_id`, `target_id` | string | endpoint claims, ordered |
| `level2_sense` | string | one of the twelve senses |
| `level1_coupling` | string | *derived*: $\mathcal{C}$(sense) |
| `directed` | bool | *derived*: whether endpoint order is meaningful |
| `ordering_only` | bool | *derived*: carries no truth coupling, produces no factor |
| `exhaustive` | bool | *derived*: true iff the coupling is EXCLUSIVE |
| `intended_strength_band` | string | strong, moderate, or weak |
| `strength_range` | [float, float] | the band's numeric interval |
| `validity` | string | valid or invalid |
| `error_kind` | string\|null | for invalid edges: wrong_sense, wrong_direction, false_endpoint, spurious |
| `is_concession` | bool | whether the relation is a conceded tension |
| `is_resolved_concession` | bool | whether the text resolves it |
| `resolver_atom_id` | string\|null | the holding claim that resolves it |
| `position_distance` | int | signed target-minus-source position gap |
| `window_admission` | string | whether the pair falls inside the locality window |
| `provenance` | object | which stage planned or transformed the edge |
| *Negative set and expectations* | | |
| `non_relations` | list | declared-unrelated claim pairs with their position distance |
| `expected.family_id`, `expected.family` | string | ladder membership and type |
| `expected.rung_index`, `expected.rung_name` | int, string | position in the ladder |
| `expected.perturbation` | object | the calls applied, their targets, their parent rung, and the declared `edge_effects` |
| `meta` | object | canonical topic, domain, framing, split, word count, generator |

world. Gold graphs are scored with $\pi_i = 0.9$ for factual claims and $0.1$ otherwise. These readouts have *different* declared expectations under C2: the consistency readout is expected to dip when a conceded tension is resolved, and the manifest asserts that dip.

**Extraction metrics.** Mined relations are matched to gold at three nested levels: **pair** (were the claims related at all), **coupling** (the Level-1 coupling, with direction for the asymmetric ones), and **sense**. Coupling is the level to report, since it is the level at which a scorer selects a factor.

Table 7: Declared ordering constraints satisfied by the MRF measure scored from the released corpus's own gold graphs, by ladder and constraint class. GOLD uses every gold edge; GOLD_VALID discards the planted-invalid ones. A dash marks a class the ladder does not declare.

| Arm | Ladder | C1 | C2 | C3 | total (%) |
|---|---|---|---|---|---|
| GOLD | CONFLICT | 339/414 | 45/92 | 119/138 | 503/644 (78.1) |
| | CHAIN | 174/189 | 62/63 | 62/63 | 298/315 (94.6) |
| | ORDER | — | 252/252 | 63/63 | 315/315 (100) |
| | CONTROL | — | 144/144 | 36/36 | 180/180 (100) |
| | all | 513/603 | 503/551 | 280/300 | **1296/1454 (89.1)** |
| GOLD_VALID | all | 277/603 | 503/551 | 238/300 | 1018/1454 (70.0) |

Recall is stratified by direction in claim order, by directedness, by locality and by validity; on the invalid subset it reads inverted, since those relations are planted errors. Precision is reported against the declared non-relations separately from precision against all unlabelled pairs, and edges outside the scheme's (sense, coupling) combinations are counted as unscorable. The ordering checker evaluates every declared constraint and records the observed direction and magnitude, so a near-miss is distinguishable from an inversion.

## 10 VALIDATING THE BENCHMARK BY MEASUREMENT

**Setup, stated first.** Two corpora are measured here, and every table states which. §10.1 reports the gold arm on the released 100-family corpus (500 responses, 1,454 declared constraints); it needs no LLM and is deterministic. The single-score baselines of §10.4 are on the release as well. The relation-extraction results of §10.2 are on the 14-family development set (70 responses, 202 constraints), on which they were run; [rerun the mined arms on the release]. Relations are mined by `llama-3.3-70b-instruct` and `gpt-oss-120b` under two candidate-pair policies: *windowed* (forward pairs within four positions, refined by what the response links) and *bidirectional* (both directions of every pair within the window). Mining is sampled once per cell; repeated runs of an identical configuration moved coupling F1 by up to 0.06, so differences smaller than that between mined arms are not interpretable. Baselines are the DiscoScore intrinsic metrics (Zhao et al., 2023), ROSCOE self-consistency (Golovneva et al., 2023), an NLI contradiction score, a G-Eval coherence judge with the SummEval rubric (Liu et al., 2023; Fabbri et al., 2021), a direct coherence judge (prompts in Appendix B), and two controls, response length and claim count.

### 10.1 GOLD-GRAPH SCORES ON THE RELEASED CORPUS

Table 7 gives the gold arm on the release. Three results matter. First, both invariance ladders are *exactly* flat: all 495 invariance and endpoint assertions on ORDER and CONTROL hold, which is what certifies that their edits change no factor. On the first assembled version of the corpus CONTROL held 96 of 180, the defect of §10.5. Second, CHAIN holds 94.6%, against 44.8% before its targeting was fixed. Third, CONFLICT holds 78.1%, and its failures are concentrated rather than diffuse. The largest block, 46 of the 47 failing C2 assertions, is the declared *invariance* of the log-partition readout at the `concession_resolved` step, which failed on every one of the 46 families, as a small decrease on 40 and a small increase on 6. That declaration was made on the pilot. Unlike the CHAIN case it is not a uniform reversal: the readout is simply not exactly invariant at this step. The natural revision is therefore to leave the pair unconstrained, not to predict a direction; [decide whether to revise it]. The next largest are C1 increases on the mean-marginal readout at the two upper steps (53), 60 of the 141 CONFLICT failures overall being changes below $10^{-3}$. GOLD_VALID falls to 70.0% because the conflict-fixing rungs remove planted-invalid conflicts that this arm has already discarded, which confirms that the planted errors carry the CONFLICT signal.

### 10.2 THE LADDERS DISCRIMINATE BETWEEN RELATION EXTRACTORS

Table 8 shows the corpus separating four extraction configurations across 22.6 F1 points. Three findings are corpus properties rather than facts about one extractor. *The declared non-relations work*

Table 8: Relation extraction against gold on the **development set**, over 603 edge-producing gold relations, micro-averaged. **Fwd/Bwd** split coupling recall by relation direction in claim order. **Non-rel** counts violations of the 342 declared non-relations.

| Extractor | Policy | Edges | Pair R | Coup P | Coup R | Coup F1 | Fwd | Bwd | Non-rel |
|---|---|---|---|---|---|---|---|---|---|
| `llama-3.3-70b` | windowed | 1106 | 82.3 | 34.1 | 62.5 | 44.1 | 71.1 | 21.9 | 34 |
| `llama-3.3-70b` | bidirectional | 3172 | 91.7 | 17.9 | 79.3 | 29.3 | 79.9 | 76.2 | 119 |
| `gpt-oss-120b` | windowed | 1094 | 83.4 | 40.2 | 73.0 | **51.9** | 82.1 | 29.5 | 38 |
| `gpt-oss-120b` | bidirectional | 2973 | 96.5 | 22.4 | 88.4 | 35.8 | 89.0 | 85.7 | 147 |

Table 9: Declared ordering constraints satisfied on the **development set**, by ladder type, for the MRF measure under each arm (202 assertions, scored against the development set's own declarations). The arms differ only in the relation graph. The development set predates the CHAIN fix of §10.5, which is why its gold-arm CHAIN column is low; compare Table 7.

| Arm | CONFLICT | CHAIN | ORDER | CONTROL | total (%) |
|---|---|---|---|---|---|
| GOLD | 89/112 | 18/30 | 45/45 | 15/15 | 167/202 (**82.7**) |
| GOLD_VALID | 37/112 | 12/30 | 45/45 | 15/15 | 109/202 (54.0) |
| MINED, `llama`, windowed | 53/112 | 17/30 | 0/45 | 0/15 | 70/202 (34.7) |
| MINED, `llama`, bidirectional | 57/112 | 18/30 | 0/45 | 0/15 | 75/202 (37.1) |
| MINED, `gpt-oss`, windowed | 59/112 | 12/30 | 0/45 | 0/15 | 71/202 (35.1) |
| MINED, `gpt-oss`, bidirectional | 53/112 | 17/30 | 0/45 | 0/15 | 70/202 (34.7) |

*as a negative set*: violations range from 34 to 147 of 342 and rank the configurations in the same order as coupling precision, while being interpretable in a way precision against unlabelled pairs is not. *Locality metadata exposes a structural limitation*: backward-running gold relations are recovered at 22–30% by the forward-only policy and 76–86% by the bidirectional one, because a forward-only candidate set cannot emit a backward directed arc; without each relation's signed position distance this would be indistinguishable from a model failure. *Recall is cheap and precision is not*: the bidirectional policy raises coupling recall by 15–17 points at the cost of tripling the edge count and quadrupling non-relation violations.

### 10.3 THE LADDERS LOCATE THE BOTTLENECK IN EXTRACTION

Table 9 gives the ordering results on the development set. The same scorer satisfies 82.7% of constraints from gold graphs and 34.7–37.1% from mined graphs; since the arms differ only in the relation graph, the gap is the extraction contribution, isolated by construction. Two features of the breakdown are informative. First, every mined arm fails *every* invariance constraint. Re-mining a reordered or re-directed response returns a slightly different graph, so the score moves; the invariance ladders therefore expose extractor *instability under meaning-preserving edits*, which no increase-only benchmark could detect. Second, GOLD_VALID falls to 54.0% because the conflict-fixing rungs remove planted-invalid conflicts that this arm has already discarded, which is expected and confirms that the planted errors carry the CONFLICT signal.

**Why the development set's gold arm is not at 100%.** Of its 35 gold-arm failures, 23 are on CONFLICT and match the release's pattern: C1 increases on the mean-marginal readout below $10^{-3}$, the log-partition invariance at `concession_resolved` coming out as a small decrease, and endpoint inversions on the consistency readout, which §9 anticipates for families whose conflicts are all EXCLUSIVE. The other 12, on CHAIN, are the targeting defect of §10.5: the breaks were not entailment breaks. We first attributed them to an over-strong declared expectation, and that diagnosis was wrong. A benchmark whose gold arm scored 202/202 would have constraints too weak to be informative.

### 10.4 THE LADDERS DISCRIMINATE BETWEEN COHERENCE MEASURES

Table 10 scores single-score measures on the same ladders. No evidence-free measure approaches the gold-graph measure on increase. The two that do best, the DiscoScore cohesion metrics at 55–59%,

Table 10: Single-score measures on the readout-independent constraints of the **released corpus**: **Increase** over the 67 CONFLICT and CHAIN families (335 assertions: C1 adjacent pairs and the C3 endpoint pair, deduplicated across readouts), **Invariance** over the 33 ORDER and CONTROL families (165 rung pairs, satisfied when the score does not move). A measure must do well on *both* columns: the claim-count control and the claim-only measures score perfectly on invariance by never moving at all. Only the two judges call the backend model, so every other row is the same under both. The MRF rows use gold graphs and no LLM. On the log-partition readout the Increase axis holds CHAIN's three breaking steps to an increase where C2 predicts a decrease (§4.1), which is most of that row's shortfall.

| | `gpt-oss-120b` | | `llama-3.3-70b` | |
|---|---|---|---|---|
| Measure | Increase | Invariance | Increase | Invariance |
| MRF, GOLD, mean marginal | 273/335 (81.5) | 165/165 (100) | (no LLM) | |
| MRF, GOLD, consistency | 242/335 (72.2) | 165/165 (100) | (no LLM) | |
| MRF, GOLD, log partition | 233/335 (69.6) | 165/165 (100) | (no LLM) | |
| DiscoScore RC | 199 (59.4) | 110 (66.7) | 199 (59.4) | 110 (66.7) |
| DiscoScore LC | 186 (55.5) | 112 (67.9) | 186 (55.5) | 112 (67.9) |
| G-Eval judge | 123 (36.7) | 62 (37.6) | 11 (3.3) | 133 (80.6) |
| Direct judge | 115 (34.3) | 48 (29.1) | 32 (9.6) | 113 (68.5) |
| Entity grid | 3 (0.9) | 153 (92.7) | 3 (0.9) | 153 (92.7) |
| NLI contradiction | 0 | 165 (100) | 0 | 165 (100) |
| ROSCOE self-consistency (max, mean) | 0 | 165 (100) | 0 | 165 (100) |
| Response length (control) | 140 (41.8) | 110 (66.7) | 140 (41.8) | 110 (66.7) |
| Claim count (control) | 0 | 165 (100) | 0 | 165 (100) |

are the only ones above the response-length control ($41.8\%$), and they measure noun repetition, which the CONFLICT and CHAIN edits happen to change; on invariance they are no better than length. The LLM coherence judges fall below the length control on increase with both backends, and fail in opposite directions: with `gpt-oss-120b` they move on most edits, meaning-preserving or not (34–37$\%$ increase, 29–38$\%$ invariance), and with `llama-3.3-70b` they barely move (3–10$\%$ increase, 69–81$\%$ invariance). The two-column design is what separates these: on either column alone a degenerate measure wins — the claim-count control on invariance, an over-sensitive judge on increase.

**Claim-only measures cannot see a ladder.** Every rung of a family carries the same claims; only the prose and the relations it draws differ. The NLI contradiction score and ROSCOE self-consistency read only the claims, so on a ladder they receive the same input at every rung, and the right result is the claim-count control's: one value per family. Scoring each distinct claim pair once per family gives exactly that. On the development set they were scored once per rung, and their rung-to-rung differences — e.g. a family's contradiction count of $14, 12, 14, 14, 14$ over five identical inputs — were sampling noise that the increase axis credited (12–17 of 50). Those development-set rows were therefore not measurements of coherence, and we do not report them. The development-set results for the mined MRF arms (§10.2) are unaffected: they read the prose, and every rung's mined graph is its own. Scoring these measures with the response as NLI context would let them see the ladder, but that is a different measure from the one the literature reports.

### 10.5 CONSTRUCTION DEFECTS

Eight generator defects passed every content gate. Seven were caught only because the gold arm makes a ladder's *internal* consistency measurable; the eighth, which the gold arm cannot see, was caught by auditing the released text against its labels. We report them because any benchmark that grades comparisons rather than absolute values is exposed to the same class of defect, and because four of them were present while earlier numbers in this line of work were reported.

**Found on the development set.** *Every rung carried the base graph.* An earlier generator perturbed each rung's text but re-attached the base plan's relation graph to all five rungs, so every gold-arm score was identical across a family. On the two families then generated, gold-arm satisfaction was

$2/28$; deriving each rung's graph from its own operator calls raised it to $22/28$. *The three shuffle rungs were identical*: the ORDER ladder's three granularities were rendered to P5 as the same call. *Plans lacked droppable conflict edges*: the deepest CONFLICT rung needs three distinct unresolved conflict edges, which nothing in P3 required.

**Found while building the release.** *An operator the prompt never defined.* Every CONFLICT ladder dispatches add_resolution on three rungs, but P5 defined only ten of the eleven calls. `claude-opus-5` inferred the operator from its name, so the development set never showed the gap. `claude-opus-5-5` returned the text unchanged or returned a comment, and 17 of the first 31 CONFLICT families were rejected. A test now asserts that P5 defines every call a ladder dispatches.

*The control ladder reversed directed edges.* direction_reversal targeted the first edge in the plan, whatever its sense. On 7 of 12 CONTROL families that was a Concession, Contrast, Condition or Cause-Effect edge, whose reversal changes a factor, so the ladder built to be flat was not: its gold arm held 96 of 180 assertions. The development set's single CONTROL family began with a Precedence edge, which is why it passed. Eligibility is now derived from the factor tables (§5.6), and the regenerated ladder holds $180/180$.

*The chain ladder did not break chains.* break_chain had the same fall-through: of its 69 targets in the first assembled release, only 10 were entailment edges. The rest were Precedence, Restatement, Alternative, Disjunction or Contrast, so a "broken" rung often lost a conflict and became *more* coherent. No CHAIN family in either corpus broke only chain links, and plans could not supply three valid entailments, because P3 never asked for a chain. The fix needed both a ladder-specific P3 quota (§5) and restricted targeting. The 21 CHAIN families were re-planned from P3, and the ladder's gold arm went from $44.8\%$ to $94.6\%$.

*Identical text under different labels.* One CHAIN family's `break_1` and `coherent` rungs came back byte-identical while declaring different edits. The adjacency gate compares edge sets between neighbours only, so it could not see this. The text–label gate of §5.6 now does, and on its first run over the corpus it also flagged a CONTROL family whose two edits had landed on the same edge.

**Found auditing the release.** *False claims corrected to the truth.* P4 must assert every planned claim, including the planted-false ones, but in 4 admitted families (one CONTROL, three CHAIN) the generator had replaced *every* false claim with its true counterpart — 9 of the 204 false claims in the release, e.g. a court of inquiry said to have blamed a bunker fire, in prose stating it "did not adopt the coal-fire explanation". Those responses carried no false content while their labels said otherwise. V4 had flagged each such claim altered by a majority, but altered did not gate. It now does on a planted-false claim; replayed on the stored votes the rule rejects exactly these 4 families and no other. They were regenerated from P1 (two needed a second attempt, after a prose-audit rejection), and their new false claims are each asserted as written. This defect is invisible to the gold arm, which reads only the labels: all four families scored $15/15$ before and after, so the gold-arm totals are unchanged. It is the one defect here that would have corrupted extraction-arm results rather than the ladder.

**A declaration revised on measurement.** With break_chain fixed, the log-partition readout fell strictly across CHAIN's three breaking steps on all 16 families first regenerated: 0 of 48 declared increases held, while the mean-marginal readout rose on all 16. The fall follows from the readout's construction (§4.1), and we moved those three pairs from C1 to a C2 predicted decrease. This revision was made *after* measuring, on 16 of the 21 families whose results it then governs. We report it as such: the evidence for the declaration is the measurement it explains, and an independent test of it requires families generated after it was fixed. The mean-marginal and consistency declarations, and the log-partition endpoint separation, were not revised.

In every case each individual item was well formed, and only the relationship between items was degenerate.

## 11 VALIDATING THE BENCHMARK WITH PEOPLE

The measurements of §10 establish that the benchmark discriminates. They cannot establish that its labels are correct, because every label and every acceptance decision is model output. This section

Table 11: Pilot human study: 4 annotators, 14 screens, 56 ratings. Agreement is Krippendorff's $\alpha$ (nominal) over the choices $\{A, B, \text{equal}\}$. Two increase screens had no majority.

| Screen kind | Screens | Ratings | Outcome |
|---|---|---|---|
| Increase (CONFLICT, CHAIN) | 10 | 40 | majority matches declared on $8/10$; ratings $32/40$ |
| Invariance, CONTROL | 2 | 8 | "equal" on $6/8$ ratings |
| Invariance, ORDER | 2 | 8 | "equal" on $0/8$; unshuffled preferred on $8/8$ |
| Agreement | 14 | 56 | $\alpha = 0.75$ overall, $0.74$ increase, $0.67$ invariance |

reports a pilot study and specifies the validation study for the release. Four claims the benchmark makes could reasonably be doubted:

**A. Label validity.** The gold graph of a base response describes that response.

**B. Edit fidelity.** Each rung's edit does what its record says, and nothing else.

**C. Ordering validity.** Readers perceive the declared orderings, and the invariances where they are declared.

**D. Committee validity.** The committee's gates agree with human judgement well enough to stand in for it.

The validation study tests the first three. Every screen is built from redacted items from which each answer-revealing field has been removed, with the side of each pair randomized; the redaction is asserted against the serialized screens rather than assumed.

### 11.1 PILOT STUDY

The pilot tested Claim C on the current release. Four annotators judged 14 pairwise screens drawn from seven families, chosen where the gold-graph measure and the strongest evidence-free baseline disagree most: CONFLICT and CHAIN pairs at the hardest adjacent step $(\rho_0, \rho_1)$ and at the endpoints $(\rho_0, \rho_4)$, and two pairs each from one ORDER and one CONTROL family. Each screen asked which response is more *logically coherent* — whether everything it says can hold together — with an explicit *about equal* option, a confidence rating and a one-sentence reason. The written brief stated that length, fluency, vocabulary, topic knowledge and sentence order in itself are not the target.

Table 11 gives the results. On increase screens readers agreed with the declared ordering on 32 of 40 ratings (one-sided binomial against a fair coin, $p < 10^{-4}$), and every annotator individually agreed on at least 7 of 10. Both endpoint and adjacent pairs were resolved, but the one screen on which all four readers answered *equal* was an adjacent CONFLICT pair $(\rho_0, \rho_1)$ whose spurious relation none of them noticed — the expected location of the benchmark's subtlest contrast. Of the 28 reasons on the increase screens that could be checked, 13 named a claim the perturbation record says was edited.

The invariance result splits by ladder type (Fisher's exact test on the $2 \times 2$, $p = 0.007$). Readers called CONTROL rungs equal, consistent with the declared invariance. They unanimously preferred the unshuffled ORDER rung, and their reasons cite stranded references ("cites *that deficit* before any deficit has been introduced") — a failure of *discourse* coherence, which the brief told them to disregard but which evidently cannot be disregarded. Two readings are possible, and the instruction test of the validation study (§11.2) is designed to choose between them. Either readers cannot separate logical from discourse coherence when the latter is degraded, in which case the ORDER invariance is a property of logical coherence that human judgement does not directly measure; or the shuffle genuinely damages the argument, in which case the declared pattern is wrong. Until that is resolved we report ORDER results separately from CONTROL and do not pool them as "invariance".

The pilot has clear limits: seven families, one pair per screen type, four annotators, one instruction, and no test of Claims A, B or D. The validation study addresses the first three.

### 11.2 VALIDATION STUDY

The study tests Claims A, B and C in a single annotation design that keeps every annotator under three hours. Claim D is deferred (§11.2). The design rests on two measurements of the release. First,

an edited rung is a small change to its base: a CONFLICT, CHAIN or CONTROL rung differs from its parent by a median of 37 words (90th percentile 75), against 643 words for a whole response. A reader who has just read a base response can therefore judge each of its rungs from the highlighted change, without reading the rung in full. Second, 62 of the 63 ORDER shuffle rungs are exact sentence permutations of their base, which a script certifies without a reader; the exception, rung 0 of f063, rewrites two sentences and is judged by hand. The study therefore reads each base response once and asks every question about its family while the text is fresh. Reading the same responses separately for labels, for edits and for pairwise comparison — as a design with one study per claim does — costs about 550 annotator-hours at the release's scale; this design costs about 30.

**Sample.** 27 families, drawn per ladder type with a fixed seed: 11 CONFLICT and 6 CHAIN, in proportion to the release; 5 CONTROL, oversampled because the ladder is small and Hypothesis H3 depends on it; and 5 ORDER, used only in the instruction test below. The 22 non-ORDER base responses carry 162 valid and 108 planted-invalid gold relations, 135 declared non-relations and 55 planted-false claims. Every sense the scheme exists to separate is represented — Contrast 43, Alternative 28, Disjunction 27, Concession 28 — but Instantiation (6) and Condition (3) are too rare to estimate alone, and are reported pooled with the other senses of their coupling.

**Core sessions.** Seven annotators with a linguistics or NLP background each judge 7 or 8 CONFLICT or CHAIN families and 2 or 3 CONTROL families, so that each of the 22 families is judged by exactly three annotators, and no annotator sees two rungs of one family in different sessions. For each family the annotator works through four steps, never seeing validity labels, rung names, or which version is the base:

1. *Read* the base response, with its claims numbered in the margin.
2. *Check* 15 *claim pairs*: the 10 gold relations and 5 declared non-relations, shuffled and unlabelled. For each: does the response draw a link between the two claims (yes / no); and if so, can both claims be true together, and can both be false together? The two truth questions recover the coupling from first principles rather than from a sense name — both-false rules out EXCLUSIVE and CO_NECESSITY, both-true rules out CONTRADICTION and EXCLUSIVE — so a reader who does not know the sense inventory is still scored correctly.
3. *Check the claims as stated*: for each numbered claim, does the response assert it as written? The planted-false claims are among them, unmarked. This tests the defect of §10.5 in which the generator corrected false claims to the truth, and costs about a minute because the claims were just read.
4. *Judge each edited version* — four for CONFLICT and CHAIN, two for CONTROL — shown as the base with its change highlighted. Is the response now more coherent, less coherent, or about as coherent; and did the change alter anything other than a link between claims (none / minor / material, with a quoted span)?

Step 2 measures Claim A, step 3 its factuality part, and step 4 Claims B and C at once. Each session begins with a brief and one practice family, excluded from analysis, whose gold answers are adjudicated in advance.

**Instruction test.** The pilot's ORDER result has two readings (§11.1), and choosing between them needs whole texts but no label work, so it runs as a separate block. Three further annotators — educated readers, screened for reading comprehension — each judge 20 whole-text pairs: the 5 ORDER families' four shuffled rungs, each against its base. The instruction varies by screen, randomized and balanced within each annotator: half the screens ask which response is more *logically consistent* ("if you believed everything each response says, which would leave you believing something consistent?"), and half which *hangs together better as an argument*. Varying the instruction within rather than between annotators makes each annotator their own control, so differences in strictness cancel, and three annotators yield 30 judgements per instruction.

**Time.** At 200 words per minute, 27 seconds per claim pair and 45 seconds per edited version, a CONFLICT or CHAIN family takes about 15 minutes and a CONTROL family about 13. The longest core session is modelled at 169 minutes including a ten-minute break, and the others at 154–167; an instruction-test session at 164 minutes, at about seven minutes per pair of full readings. These are

Table 12: Hypotheses of the validation study. *Judged* counts judgements over all three raters; *Units* counts the items each test is run on, one majority per item; *Precision* is the 95% Wilson half-width on those units at the expected rate, or what the test needs for power 0.8.

| Hypothesis | Judged | Units | Precision | Pass criterion |
|---|---|---|---|---|
| A1. Valid gold relations are confirmed | 486 | 162 | ±4.7 | $\geq 90\%$ at the coupling level |
| A2. Declared non-relations are confirmed | 405 | 135 | ±6.7 | $\geq 80\%$ |
| A3. Planted-invalid relations are rejected | 324 | 108 | ±7.5 | by error kind; an accepted planted error is a label to repair |
| A4. Planted-false claims are asserted | 165 | 55 | ±6.3 | $\geq$ 95%, every exception listed |
| B. Edits realize their declared effect only | 234 | 78 | ±6.8 | $\geq$ 90%, no material collateral change |
| H1. Readers agree with the declared increase | 204 | 68 | needs 23 | majority above chance, one-sided binomial |
| H2. Agreement is lowest at CONFLICT's first step | 33 | 11 | — | descriptive, with its interval |
| H3. CONTROL versions read as equal | 30 | 10 | — | descriptive |
| H4. The instruction moves ORDER toward equal | 60 | 30/arm | needs 21/arm | higher "equal" rate under logical consistency |

model estimates, not measurements. The pilot's recorded times cannot calibrate them: the annotation tool logged the time a task was open rather than the time spent on it, and one screen recorded 42 hours. Two annotators are therefore timed on the practice family before the study runs; if it takes more than 17 minutes, an eighth core annotator is added and the same 66 annotator–family slots are spread thinner, which keeps three annotators on every family. A session that reaches three hours stops at a family boundary, and its unjudged families move to the next annotator. Two attention checks are folded into each session at no cost: one claim pair repeats a claim, and one edited version restores its base word for word. Annotators who fail either are excluded before analysis.

**Hypotheses and analysis, pre-registered.** Table 12 lists the hypotheses. Each test runs on majorities, one per item, so its sample is a third of the judgements. The rate estimates A1–A4 and B are reported with 95% Wilson intervals, of half-width 4.7 to 7.5 points at the expected rates; that resolves a 90% pass threshold from a rate of 80%, but not from 85%, and the criteria are set accordingly. H1 needs 23 items for power 0.8 at $\alpha = 0.05$ at the pilot's measured agreement of 0.75, and has 68. H4 needs 21 per instruction to detect a rate of "about equal" of 0.10 against 0.40, and has 30. H2 and H3 are too small to test, and are reported descriptively. Agreement is Krippendorff's $\alpha$ per question (Krippendorff, 2018), with $\alpha \geq 0.67$ on coupling as the reliability target. Rung strengths per family are fitted with a Bradley–Terry model with ties (Bradley & Terry, 1952) and compared with the declared order by Kendall's $\tau$, and every measure of §10 is rescored against the human majority as well as against the declared answer. A rung whose edit is judged unrealized, or whose collateral change is material, has a gold graph that does not describe its text: such rungs are flagged in the release, and every result is re-reported without their families.

**Both outcomes of H4 are publishable, and their consequences are fixed in advance.** If H4 holds, the ORDER ladder is validated as an invariance test of *logical* coherence, and the benchmark documents that human "coherence" ratings conflate the two properties. If it fails, ORDER is redesignated a *discourse-coherence probe*: its declared pattern is changed to increasing toward the unshuffled base, it is removed from the invariance score, and the change is reported. Either way, ORDER and CONTROL are never pooled.

**Scope.** Claim D — that the committee's gates agree with human judgement — is not tested. Testing an admission gate needs responses the gate rejected, and the release keeps the rejection records but only two rejected responses; the rest were overwritten by the retries that replaced them. A test of Claim D therefore needs a separate generation run with the gates logged but not enforced. It would

validate the generator rather than the released labels, so we leave it to future work. Until then, the committee's verdicts are evidence that the prose realizes its plan, not a substitute for the human checks above.

**Cost and procedure.** The core sessions total about 19 annotator-hours, the instruction test about 8, and adjudicating the practice family about 3: about 30 hours in all, paid at or above the local minimum wage for the time worked. Screens are exported from the redacted release by the script of §8, which adds a highlighted-change view to the existing pairwise screen; its leak checks apply unchanged. The analysis plan — hypotheses, models, exclusion rules and pass criteria — is registered before data collection, and the pilot's four annotators do not take part. [Ethics review: obtain and report the institutional determination before data collection.]

## 12 LIMITATIONS

**The corpus is LLM-generated, including its labels.** Questions, claims, plans, responses, perturbations and gold labels are model output, validated by a model committee. The structural mitigations of §6 address the most obvious circularity, but not a bias shared between generator and committee; current frontier models are not independent in the relevant sense, and in our release one committee member shares the generator's vendor. Until Studies A, B and D are complete, the gold graphs should be treated as model-certified rather than human-verified.

**Perturbed rungs are validated by construction.** The committee checks base responses; rung gold graphs are derived by declared edge effects, and whether each P5 edit realizes its effect and nothing more is checked only by deterministic gates until the validation study's edit check (§11.2) is complete.

**Declared orderings are premises, and two were revised.** A ladder's constraints encode our judgement that, for example, resolving a conceded tension increases coherence. The pilot shows that at least one declared pattern — ORDER invariance — is not what readers perceive. The CHAIN log-partition declaration was revised after measuring the families it governs (§10.5), and the CONFLICT log-partition invariance at `concession_resolved` fails on all 46 families (40 as decreases), so it is a candidate for the same treatment. A declaration revised on the data it scores is weaker evidence than one made in advance. The validation study tests the premises (Hypotheses H1–H4 of §11.2), and we fix in advance how its outcomes change them.

**Scale, and where the extraction results come from.** One hundred families is enough to establish that the instrument discriminates, and not enough to rank closely matched measures. Mining is single-sample, with measured run-to-run variation of 0.06 F1. The extraction comparisons are on the 14-family development set, which predates the CHAIN fix, and have not been rerun on the release; the baseline comparisons are on the release. CONTROL is the smallest ladder type by design (12 families), and the type a reader should most want enlarged.

**Genre, length, and provenance.** Every item is 567–731 words of formal expository prose with 15–16 claims, and the locality bias of P3 under-represents long-range dependencies; conclusions do not transfer automatically to dialogue, summarization, much longer arguments, or informal registers. The release comes from a single generator and the mined arms from two extraction models, so generator-specific artifacts are controlled only by the V3 leakage gate. The model gateway caches identical requests, so a repeated temperature-zero run through it can return a cached response rather than a new sample. Four families were regenerated from P1 with the cache bypassed, and one release family takes its question and claims from a cached replay of a pilot run on the same slot (its plan and prose were regenerated).

## 13 CONCLUSION

Evaluating a coherence measure requires contrasts that vary coherence while holding content fixed. LoCoBench supplies them as ladders: families of responses over one claim set, with declared orderings, per-rung gold relation graphs, a labelled negative set, and invariance ladders on which a

sound measure must not move. The construction attributes a score difference to coherence rather than to content, and — by scoring a measure once on gold graphs and once on mined ones — isolates extraction error from scoring error by construction. On the released $100$-family corpus the gold-graph measure satisfies $89.1\%$ of $1{,}454$ declared constraints, with both invariance ladders exactly flat. On the development set, the same scorer satisfies $82.7\%$ from gold graphs and at most $37.1\%$ from mined ones, and mined graphs are unstable under meaning-preserving edits: the bottleneck is extraction. On the release, only two cohesion metrics beat the response-length control on increase, LLM judges fall below it with both backends, and measures that read only the claims cannot see a ladder at all.

The gold arm is also how the benchmark audits itself. Seven of the eight construction defects that passed every content gate were found this way, including a CHAIN ladder that had never broken a chain and a CONTROL ladder that was not meaning-preserving; each was visible only as a gold-arm result the construction should not have produced. A pilot study finds that readers confirm the declared orderings and the meaning-preserving control, and reject the invariance of sentence shuffling; the validation study we specify will establish how far the labels and the edits can be trusted.

## AI USE STATEMENT

This is a benchmark constructed *by* generative AI, and the disclosure is correspondingly extensive. Every artifact in the corpus — questions, claims, relation plans, response prose, perturbed rungs and gold relation labels — is the output of a large language model driven by the prompts of Appendix A, and every acceptance decision is made by a committee of three further language models. §11 and §12 state what this does and does not establish. We also used AI assistants for code implementation and editorial assistance on this manuscript. All LLM-generated code was reviewed by the authors and is covered by a test suite that, among other things, asserts every prompt in the appendix byte-for-byte against the code. The benchmark design — the ladder construction, the constraint classes, the operator inventory, the gate thresholds and their calibration, the schema, the metrics and the human-study protocol — is the authors' own. We take responsibility for the final content of this work.

## ETHICS STATEMENT

**Intended use.** LoCoBench is for evaluating measures of logical coherence in long-form generated text. It is not a factuality benchmark: a response can be perfectly coherent and entirely false. It is not a training corpus for text generation.

**Deliberately false content.** Every item asserts at least two factually false claims, and every plan contains four deliberately invalid relations. This is essential to the design and makes the corpus hazardous if mishandled. False claims are labelled at the claim level and invalid relations at the edge level, and **the corpus must not be redistributed with those labels stripped**. Prose from these items should not be treated as a source of factual information.

**Human participants.** The pilot study of §11.1 involved four volunteer annotators who gave informed consent, judged generated text only, and provided no personal data beyond an anonymous identifier. The validation study of §11.2 will recruit ten paid annotators under an institutional ethics determination obtained before data collection, with payment at or above the local minimum wage for the time worked, and no session longer than three hours. No item concerns a real case or an identifiable individual.

**Compute.** Generating one family takes several dozen model calls across generation and validation, and a mined arm over the corpus several thousand more. This is modest but not free, and a benchmark that requires an LLM committee to extend is less accessible than one that does not.

## REPRODUCIBILITY STATEMENT

Appendix A reproduces every prompt the pipeline, the committee, the retry mechanism, the relation miner and the judge baselines issue; the files are written from the source by a script and a test fails if they diverge. Appendix C gives the prompt specification and substitution contract. Table 4 gives every acceptance threshold, Table 3 the perturbation operators, Table 6 the item schema, Algorithm 1 the

procedure, and §5.6 the derivation of each rung's gold graph. §6 names the generating and committee models. The release comprises the items, the family manifests with their declared constraints, the generation log with each admitted family's committee verdict, and the rejected families with their reasons. The gold-arm evaluation requires no LLM access and is deterministic.

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# A THE GENERATION AND VALIDATION PROMPTS, VERBATIM

Every prompt follows, exactly as issued, written to the files included here by `scripts/export_locobench_prompts.py` and checked against the code by a test. Placeholders have the form `[NAME_PLACEHOLDER]` and are substituted by the contract of Appendix C. Line numbers are added for reference. Each prompt is preceded by a short card: its inputs, what it must emit, and what checks its output.

## A.1 P1 — TOPIC TO QUESTION

**Input:** a canonical topic. **Output:** one open-ended question. **Checked by:** the parser (a single question). 1,902 characters, 9 numbered instructions. Placeholder: `[TOPIC_PLACEHOLDER]`.

```
Instructions:
1. You are given a TOPIC. Generate one open-ended QUESTION about that topic.
2. The QUESTION must begin with Why, How, What, Does, or Which.
3. The QUESTION must elicit an answer containing MULTIPLE, DIVERSE logical
   relationships. Aim for an answer that would naturally contain at least four
   of the following: a causal chain; a condition; a temporal ordering; an
   evidential appeal; a general claim with a specific instance; a restatement
   or definitional equivalence; a non-exhaustive contrast; an EXHAUSTIVE pair
   of competing alternatives of which EXACTLY ONE can hold; a disjunctive pair
   of which AT LEAST ONE must hold; a tension that an authoritative finding
   later resolves.
4. Prefer topics where competing explanations are adjudicated: incident
   investigations, litigation, diagnostic reasoning, attribution disputes,
   competing scientific hypotheses. Such topics naturally produce exhaustive
   alternatives and resolving holdings.
5. The QUESTION must be answerable in 500-900 words of formal prose.
6. The QUESTION must not itself contain the words "cause", "condition",
   "alternative", or "contrast", so the answer's relations are not primed
   lexically.
7. Do NOT ask a multi-part question joined by "and" - one question only.
8. Wrap the QUESTION in square brackets.
9. Output the bracketed QUESTION only, with no preamble.

Use the following examples to better understand your task.

TOPIC: aviation component failure investigation
QUESTION: [How do investigators determine responsibility when a component
supplied by a third party fails in service and the operator and the
manufacturer offer competing explanations?]

TOPIC: antibiotic resistance in hospital settings
QUESTION: [Why does resistance emerge in some hospital units but not others,
and how is a specific outbreak traced to its origin?]

Your Task:
TOPIC: [TOPIC_PLACEHOLDER]
QUESTION:
```

## A.2 P2 — QUESTION TO TAGGED CLAIMS

**Input:** the P1 question. **Output:** 26 atomic claims, each tagged correct, incorrect, alt-pair, disj-pair, equiv-pair or holding. **Checked by:** the parser (count and tag composition). 3,752 characters, 7 numbered instructions. Placeholder: `[QUESTION_PLACEHOLDER]`.

```
Instructions:
1. You are given a QUESTION. Generate a set of ATOMIC CLAIMS that a
   well-informed answer to it might contain.
2. An atomic claim is a sentence containing a singular piece of information,
   which should be basic, indivisible, and independently verifiable.
3. Generate exactly 26 atomic claims: the 24 below, plus the 2 holding claims
   required by instruction 4. The 24 break down as:
   (a) 14 factually correct claims;
   (b) 4 factually incorrect claims (plausible but false);
   (c) 2 EXHAUSTIVE ALTERNATIVE claims: a pair X, Y such that exactly one of X
       and Y must be true - they cannot both hold, and they cannot both fail.
       Example pair: "No one was harmed in the incident." / "Three people died
       in the incident."
   (d) 2 DISJUNCTIVE claims: a pair X, Y such that at least one must be true,
       but both may be true simultaneously. Make each one a STATE OF THE WORLD,
       not a finding or a check that reported something: "the alloy was
       substituted" and "the batch was mislabelled", not "assay A flagged it"
```

```
      and "assay B flagged it".
   (e) 2 EQUIVALENT claims: a pair X, Y asserting the same fact in different
      words, so that either one being true makes the other true.
4. Additionally generate 2 HOLDING claims: sentences in which an authority
   (a court, tribunal, regulator, final report, or review board) reaches a
   determination that settles a dispute between two other claims in your list.
   A holding claim must use adjudicating language - "held", "ruled",
   "concluded", "determined", "found that", "ultimately".
5. Each claim must be a single declarative sentence with no connective linking
   it to another claim. Do NOT use "because", "although", "therefore", "if",
   "while", "however". Relations are added at a later stage, not here.
6. Do NOT indicate which claims are factually incorrect in their wording; do
   NOT hedge with "assume", "might", "possibly", or "allegedly".
7. Output as a markdown code block: an unordered list with "-" bullets. Append
   a tag in square brackets to every claim, one of: [correct], [incorrect],
   [alt-pair-1], [alt-pair-2], [disj-pair-1], [disj-pair-2], [equiv-pair-1],
   [equiv-pair-2], [holding].

Use the following example to better understand your task.

QUESTION: [How do investigators determine responsibility when a component
supplied by a third party fails in service and the operator and the
manufacturer offer competing explanations?]

ATOMIC CLAIMS:
```
- AeroParts supplied turbine blades to the carrier. [correct]
- The turbine blades were installed in the carrier's A-320 fleet. [correct]
- Seventeen turbine blades failed in flight during service. [correct]
- The regulator opened a safety probe into the blade failures. [correct]
- Maintenance logs recorded abnormal vibration on the failed blades. [correct]
- The airline argued that the failures were caused by pilot error. [correct]
- The final report concluded the root cause was a metallurgical defect. [correct]
- The regulator issued an order grounding the fleet. [correct]
- Every affected aircraft was pulled from service. [equiv-pair-1]
- The regulator's order removed all affected aircraft from operation. [equiv-pair-2]
- The blades were certified under a 1998 airworthiness standard. [incorrect]
- No one was harmed in the incidents. [alt-pair-1]
- Three passengers died in one of the failures. [alt-pair-2]
- The alloy supplied for the blades was substituted. [disj-pair-1]
- The blade batch was mislabelled at the mill. [disj-pair-2]
- The tribunal held that AeroParts was liable for the blade failures. [holding]
- The accident report found the pilots not at fault. [holding]
```

Your Task:
QUESTION: [QUESTION_PLACEHOLDER]
ATOMIC CLAIMS:
```

### A.3 P3 — CLAIMS TO RELATION PLAN

**Input:** the question and the tagged claims. **Output:** a JSON plan of 14–16 positioned claims, ten relations with sense, strength band, validity and error kind, and 4–6 non-relations. **Checked by:** the plan gate (every P3 row of Table 4, the required senses, the conflict-sense quota, disjointness of the relation and non-relation lists), whose complaint is fed back as a content retry. 7,778 characters, 10 numbered instructions. Placeholders: `[QUESTION_PLACEHOLDER]`, `[CLAIMS_PLACEHOLDER]`.

```
Instructions:
1. You are given a QUESTION and a tagged list of ATOMIC CLAIMS. Produce a
   RELATION PLAN: a machine-readable graph of the logical relations a response
   should draw between those claims.
2. Select 14-16 claims to use. Aim for 15. Number them with FRESH consecutive
   positions: the first claim you assert is pos 1, the next pos 2, and so on
   with NO gaps and NO repeats, ending at pos N where N is how many you
   selected. Do NOT carry over the numbering of the input list - if you pick the
   3rd, 7th and 12th claims they become pos 1, 2 and 3. The position is the
   order the response will assert the claim in. Every selected claim keeps its
   text EXACTLY as given. Do NOT introduce information not in the claims, and
   do NOT alter the claims. Your selection MUST INCLUDE AT LEAST 2 of the
   [incorrect] claims: the response has to assert false content alongside the
   true, so a selection drawn only from the [correct] claims is rejected. Carry
   each selected claim's tag through as a "factual" field - false for an
   [incorrect] claim, true for every other tag.
3. Plan EXACTLY 10 RELATIONS. Each relation connects exactly two selected claims
   and has a SENSE. Every relation must join a DIFFERENT pair of claims: once two
   claims are related, that pair is used up and must not appear again in the
   relations list, in either direction. The SENSE is one of: Cause-Effect,
   Effect-Cause, Evidence, Condition,
   Restatement, Instantiation, Contrast, Concession, Alternative, Disjunction,
```

```
   Precedence, Succession.
   - Cause-Effect: source causes target. Effect-Cause: source is the effect,
     target its cause.
   - Evidence: source is evidence for target. Condition: source is a condition
     for target.
   - Restatement: source and target assert the same thing.
   - Instantiation: source is a general claim, target a specific instance.
   - Contrast: opposition that is NOT exhaustive - both could be false.
   - Concession: a tension the text itself resolves.
   - Alternative: EXHAUSTIVE competing options - EXACTLY ONE holds. Not both,
     and not neither. Use your [alt-pair-*] claims here.
   - Disjunction: AT LEAST ONE holds; both may hold. Use your [disj-pair-*]
     claims here.
   - Precedence / Succession: ordered in time with NO truth dependence -
     neither claim being true or false says anything about the other.
4. Keep relations LOCAL: prefer pairs within 4 POSITIONS of each other, so the
   graph stays recoverable from the response by a local reader. A longer-range
   pair is acceptable when the two claim texts share a named entity, and a few
   long-range links (for instance a closing claim answering the opening one) are
   fine. Order the positions so that related claims sit near each other.
5. The plan must include AT LEAST ONE relation of each of these senses:
   Alternative, Disjunction, Restatement, Concession, and one of
   Precedence/Succession. Separately, [LADDER_PLACEHOLDER]
6. Every Concession must be marked resolved or unresolved. A resolved
   Concession must name a RESOLVER: a [holding] claim, also selected and
   positioned AFTER both endpoints.
7. Give each relation an intended STRENGTH BAND: strong (0.85-1.00), moderate
   (0.60-0.84), or weak (0.35-0.59) - how strongly the source determines the
   target under that sense. Restatement relations are always strong.
8. Give each relation an intended VALIDITY: valid if the relation genuinely
   holds between the claims as stated; invalid if it does not. Of your 10
   relations, mark EXACTLY 6 "valid" and EXACTLY 4 "invalid" - count them before
   you emit, and do not compute a percentage. For each invalid relation give an
   ERROR_KIND, one of: wrong_sense, wrong_direction, false_endpoint, spurious.
9. Also list 4-6 NON-RELATIONS (aim for 5): pairs of selected claims within 4
   positions of each other between which the response must draw NO logical
   dependence. These are the distractors a relation miner must correctly label
   None. A non-relation pair must NOT also appear in your relations list, in
   either direction - the two lists are disjoint.
10. Output a single JSON object in a code block and nothing else. Below is a
    COMPLETE, correctly-shaped example - copy its structure exactly, not its
    text. It has 14 atoms numbered 1..14 with no gaps, 2 of them factually
    incorrect, 10 relations (6 valid, 4 invalid), all five required senses, and 5
    non-relations disjoint from the relations:

```json
{
  "atoms": [
    {"pos": 1, "text": "<claim text>", "factual": true},
    {"pos": 2, "text": "<claim text>", "factual": true},
    {"pos": 3, "text": "<claim text>", "factual": true},
    {"pos": 4, "text": "<claim text>", "factual": false},
    {"pos": 5, "text": "<claim text>", "factual": true},
    {"pos": 6, "text": "<claim text>", "factual": true},
    {"pos": 7, "text": "<claim text>", "factual": true},
    {"pos": 8, "text": "<claim text>", "factual": true},
    {"pos": 9, "text": "<claim text>", "factual": false},
    {"pos": 10, "text": "<claim text>", "factual": true},
    {"pos": 11, "text": "<claim text>", "factual": true},
    {"pos": 12, "text": "<claim text>", "factual": true},
    {"pos": 13, "text": "<claim text>", "factual": true},
    {"pos": 14, "text": "<claim text>", "factual": true}
  ],
  "relations": [
    {"source_pos": 1, "target_pos": 2, "sense": "Cause-Effect",
     "strength_band": "strong", "validity": "valid",
     "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 2, "target_pos": 3, "sense": "Evidence",
     "strength_band": "moderate", "validity": "valid",
     "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 3, "target_pos": 4, "sense": "Restatement",
     "strength_band": "strong", "validity": "valid",
     "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 5, "target_pos": 6, "sense": "Alternative",
     "strength_band": "strong", "validity": "valid",
     "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 7, "target_pos": 8, "sense": "Disjunction",
     "strength_band": "moderate", "validity": "valid",
     "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 9, "target_pos": 10, "sense": "Concession",
     "strength_band": "moderate", "validity": "valid",
     "error_kind": null, "resolved": true, "resolver_pos": 11},
    {"source_pos": 11, "target_pos": 12, "sense": "Precedence",
```
```

```
     "strength_band": "weak", "validity": "invalid",
     "error_kind": "wrong_sense", "resolved": null, "resolver_pos": null},
    {"source_pos": 12, "target_pos": 13, "sense": "Cause-Effect",
     "strength_band": "moderate", "validity": "invalid",
     "error_kind": "wrong_direction", "resolved": null, "resolver_pos": null},
    {"source_pos": 13, "target_pos": 14, "sense": "Contrast",
     "strength_band": "weak", "validity": "invalid",
     "error_kind": "spurious", "resolved": null, "resolver_pos": null},
    {"source_pos": 4, "target_pos": 6, "sense": "Contrast",
     "strength_band": "weak", "validity": "invalid",
     "error_kind": "false_endpoint", "resolved": null, "resolver_pos": null}
  ],
  "non_relations": [
    {"source_pos": 1, "target_pos": 4}, {"source_pos": 2, "target_pos": 5},
    {"source_pos": 6, "target_pos": 9}, {"source_pos": 8, "target_pos": 11},
    {"source_pos": 10, "target_pos": 13}
  ]
}
```

```
Your Task:
QUESTION: [QUESTION_PLACEHOLDER]
ATOMIC CLAIMS: [CLAIMS_PLACEHOLDER]
RELATION PLAN:
```

### A.4 P4 — PLAN TO RESPONSE

**Input:** the question and the plan. **Output:** 550–650 words of prose. **Checked by:** the parser (word floor), then V1, V3 and V4 on the committee, whose combined complaint is fed back as a content retry. 5,571 characters, 8 numbered instructions. Placeholders: `[QUESTION_PLACEHOLDER]`, `[PLAN_PLACEHOLDER]`.

```
Instructions:
1. You are given a QUESTION and a RELATION PLAN in JSON. Write a RESPONSE that
   answers the question and realizes the plan exactly.
2. Assert every planned atom EXACTLY ONCE, in the plan's POSITION order,
   preserving each atom's factual content. Do not restate an atom you have
   already asserted in order to attach a later relation to it - refer back with
   a pronoun or a short noun phrase instead. Name that phrase after the thing
   the atom is about, not after its status as a finding: "the primitive carpal
   morphology" or "that wrist evidence", not "the analysis's flagging of
   primitive traits"; "these two accounts" or "either possibility", not "the two
   claims in circulation". A summary phrase covering both endpoints of one
   connective is fine - "at least one of these must hold: X, or Y". You may
   adjust surface wording for fluency but must not change what is asserted.
   Quantifiers, determiners, numbers, dates, polarity and modals are CONTENT,
   not wording: keep "all", "only", "no", "some", "each", "never" and every
   figure exactly as the atom has them. "All tools were crafted by a single
   culture" may become "every one of the tools came from a single culture", but
   never "the tools were crafted by a single culture" - dropping "All" weakens
   the claim and counts as changing it.
   Never let one atom absorb another: each must stay separately true or false.
   Joining two atoms with a connective is fine and often required - "either X or
   Y" and "although X, Y" both assert two atoms and are exactly what instruction
   3 asks for.
3. Realize every planned RELATION so a careful reader could recover it from the
   text alone, using language appropriate to its sense:
   - Cause-Effect / Effect-Cause: "caused", "led to", "resulted from", "because"
   - Evidence: "indicates", "confirms", "as shown by"
   - Condition: "if", "provided that", "only when"
   - Restatement: "that is", "in other words", "equivalently"
   - Instantiation: "for example", "specifically", "in one case"
   - Contrast: "by contrast", "whereas", "on the other hand"
   - Concession: "although", "despite", "even though"
   - Alternative: make the exhaustiveness EXPLICIT - "either ... or", "one of
     these accounts must be wrong", "the two cannot both be true, and one of
     them must hold"
   - Disjunction: make the at-least-one reading EXPLICIT - "at least one of",
     "one or both", "perhaps both". Say "perhaps both", never "possibly both".
     A bare "X, or Y" is NOT enough for either of these two senses: on its own it
     leaves a reader unable to tell whether both claims could hold, which is the
     only thing separating Alternative from Disjunction. Every such pair needs its
     marker - "either X or Y" when exactly one holds, "at least one of X or Y,
     perhaps both" when both may.
   - Precedence / Succession: state the sequence EXPLICITLY in one sentence
     joining both claims - "before", "after", "subsequently", "earlier than",
     "which postdates" - while keeping the ordering purely chronological, so a
     reader sees only when things happened and infers nothing from it. Write
     "X, which predates Y" or "Y came after X", not "X and therefore Y".
```

```
    Asserting the two claims in separate sentences leaves this relation
    unrealized: an ordering a reader cannot see is the same as no ordering.
4. For each planned NON-RELATION, assert both atoms but draw no dependence
   between them. Separate them by DISTANCE and PARAGRAPHING, not by a marker:
   put other material between them, or place them in different paragraphs, and
   simply use no connective. Do NOT announce the absence of a link - phrases
   like "In an unrelated matter", "Separately", or "On a different note" read as
   stitched-together annotation rather than prose and will be marked down.
5. For a resolved Concession, assert the resolver atom after both endpoints and
   present it as settling the tension.
6. Write it as a continuous expository answer to the QUESTION - an argument a
   domain expert would publish, organized into 3-5 paragraphs, in which the
   planned relations are simply how the argument hangs together. Do NOT walk the
   plan relation by relation: a reader must not be able to tell that a plan
   exists. Restate a subject to carry an argument forward ("the radiocarbon
   date", "that attribution"), never the earlier sentence as a step ("the
   finding just reported", "the provision just stated", "the ordering just
   described"). Use precise and formal language, avoiding vague generalizations
   and rhetorical fillers. Aim for 550-650 words. The 500-word floor is a hard
   requirement checked by an automated reader, which rejects a shorter answer
   outright however good it is, so give each atom and each relation the space to
   be argued rather than compressing them into a summary.
7. Do NOT mention the relation plan, the senses, the strength bands, the
   validity labels, or the fact that some content is incorrect. Do NOT hedge
   with "assume", "might", "possibly", "allegedly", "supposedly",
   "reportedly", or "it is claimed" around planned-invalid relations - assert
   them in the same register as the valid ones. Do NOT add headings, bullet
   lists, or annotations.
8. Wrap the RESPONSE in a code block whose contents are the prose itself, as you
   would publish it. The block holds the answer, not a JSON object or any other
   payload with the prose inside a field.

Your Task:
QUESTION: [QUESTION_PLACEHOLDER]
RELATION PLAN: [PLAN_PLACEHOLDER]
RESPONSE:
```

### A.5 P5 — RESPONSE TO PERTURBED RUNG

**Input:** the current text, the plan, and one operator call with its target. **Output:** the edited text and a structured diff. **Checked by:** the text-changed and length-drift gates, then the adjacency gate over the whole family. 2,592 characters, 5 numbered instructions; instruction 3 defines all ten calls of Table 3. Placeholders: `[RESPONSE_PLACEHOLDER]`, `[PLAN_PLACEHOLDER]`, `[OPERATOR_PLACEHOLDER]`.

```
Instructions:
1. You are given a RESPONSE, its RELATION PLAN, and one PERTURBATION to apply.
2. Apply ONLY the named perturbation. Change the minimum number of words
   needed. Every other relation, atom, and sentence must be preserved verbatim.
3. The perturbations are defined as follows.
   - wrong_sense(edge, new_sense): rewrite the connective realizing that edge so
     the text draws new_sense instead. Keep both atoms' content unchanged.
   - direction_reversal(edge): rewrite so the relation runs from the target to
     the source. Keep both atoms' content unchanged.
   - exhaustiveness_flip(edge): if the edge is Alternative, rewrite so the two
     claims merely oppose without being exhaustive (both could be false). If it
     is Contrast, rewrite so exactly one must hold.
   - remove_resolution(edge): delete the resolver sentence and any reference to
     it, leaving the conceded tension unresolved. Change nothing else.
   - add_resolution(edge): the edge is a conflict the response leaves open. Add
     ONE sentence, after both of its atoms, in which an authority reaches a
     determination that settles the tension. Do not delete either atom or change
     the connective realizing the edge. Change nothing else.
   - inject_contradiction(atom): add ONE new sentence asserting a claim that
     directly contradicts the named atom. Place it at least two sentences later.
   - break_chain(edge): remove or negate the link realizing that edge so the
     causal chain no longer connects, keeping both endpoint atoms present.
   - shuffle_order(level): reorder the response's sentences without changing any
     wording. level=full permutes all sentences; level=block permutes paragraph
     blocks; level=adjacent swaps one adjacent pair.
   - spurious_relation(atom_a, atom_b): add a connective asserting a causal or
     contradictory link between two atoms the original response kept separate.
     Add no new factual content.
   - drop_relation(edge): remove the connective realizing that edge so the two
     atoms are merely adjacent assertions. Keep both atoms.
   - ordering_only(edge): rewrite a Precedence edge as Succession, or the
     reverse, preserving the actual chronology and adding NO causal or
     inferential language. The truth of neither atom may become dependent on
     the other.
```

```
4. Do not change the response's length by more than 15 percent, except for
   inject_contradiction / add_resolution (which add one sentence) and
   remove_resolution / break_chain (which remove one).
5. Output the perturbed RESPONSE in a code block, then a second code block
   containing a JSON diff:
   {"operator": "...", "target": ..., "sentences_changed": [...],
    "relations_added": [...], "relations_removed": [...],
    "relations_relabeled": [...]}

Your Task:
RESPONSE: [RESPONSE_PLACEHOLDER]
RELATION PLAN: [PLAN_PLACEHOLDER]
PERTURBATION: [OPERATOR_PLACEHOLDER]
```

### A.6 V1 — RELATION RECOVERY

**Input:** the response and its claims keyed $1..N$ — *not* the plan. **Output:** the relations the response draws, with sense, coupling, strength band and resolved flag. **Decision:** any-of over the committee, thresholds of Table 4. 4,086 characters. Placeholders: `[RESPONSE_PLACEHOLDER]`, `[ATOMS_PLACEHOLDER]`. The validators use a function-signature idiom rather than numbered instructions. The final rule states the compile map, so a validator's coupling must be the one its sense compiles to.

```
Let's define a function named recover_relations(response: str,
atoms: dict[str, str]).
Given a RESPONSE and the ATOMS drawn from it, the function returns the
list of logical relations the response ITSELF draws between those atoms.

ATOMS maps an atom's NUMBER to its text, numbered from 1. Use those numbers, as
integers, to identify atoms in your output: if the relation holds between the
atoms numbered "3" and "7", emit source 3 and target 7. Never renumber them and
never count from zero.

For each ORDERED pair (i, j) that the response actually relates, emit one object:
  {"source": i, "target": j,
   "sense": one of ["Cause-Effect","Effect-Cause","Evidence","Condition",
     "Restatement","Instantiation","Contrast","Concession","Alternative",
     "Disjunction","Precedence","Succession"],
   "coupling": one of ["entailment","contradiction","equivalence","exclusive",
     "co_necessity","none"],
   "strength_band": one of ["strong","moderate","weak"],
   "resolved": true or false or null}

Rules:
- Emit a relation ONLY if the response itself draws that connection, as written
  or as a clear step in the author's argument. Do not emit a relation that is
  merely plausible in general.
- Pick the MOST SPECIFIC sense the text supports, never the most general one.
  Five senses compile to "entailment", and "Evidence" is the broadest of them,
  so it is the easy wrong answer: use it only for evidential support ("indicates",
  "confirms", "as shown by", "suggests"). Prefer "Cause-Effect" when the text says
  one thing brought the other about ("caused", "led to", "resulted in",
  "because"), "Effect-Cause" when that runs the other way ("resulted from",
  "is explained by"), "Condition" for "if", "provided that", "only when", and
  "Instantiation" when one claim is an INSTANCE of the other, i.e. a specific
  case of a general statement ("for example", "specifically", "such as").
- Use "Alternative" only when the two claims cannot both be true AND cannot both
  be false. Use "Contrast" when they merely cannot both be true - look for "by
  contrast", "whereas", "on the other hand", or two claims set against each other
  without either being conceded.
- Use "Concession" when the text grants a tension and then overrides it -
  "although", "despite", "even though", "notwithstanding", or a clause that admits
  one claim while asserting the other anyway. A concessive marker attaching two
  claims IS a relation between them, even when the concession sits in a
  subordinate clause rather than its own sentence.
- Use "Disjunction" only when the response rules out both claims being false
  ("at least one of", "one or both", "perhaps both").
- Use "Precedence" or "Succession" only when the ordering carries no truth
  dependence; set their coupling to "none". These ARE relations and must be
  emitted: "before", "after", "subsequently", "earlier", "later", or an explicit
  statement that a sequence carries no causal or inferential force. Read such a
  pair as "Precedence" when the source comes first in time and "Succession" when
  it comes second. Do not silently upgrade one to "Cause-Effect" or "Evidence"
  because the two events sit next to each other in an argument.
- Relate a pair only ONCE unless the response genuinely draws two different
  connections between them. Emitting many relations does not make the recovery
  more complete; a pair the response merely mentions in the same paragraph is not
  related.
```

```
- "coupling" is not an independent choice: it must be the coupling that the
  "sense" you chose compiles to. Cause-Effect, Effect-Cause, Evidence,
  Condition and Instantiation are "entailment"; Restatement is "equivalence";
  Contrast and Concession are "contradiction"; Alternative is "exclusive";
  Disjunction is "co_necessity"; Precedence and Succession are "none".
- Pairs the response leaves unrelated must be omitted entirely.

The function returns a JSON list. Note that your response will be passed to the
python interpreter, SO NO OTHER WORDS!

recover_relations(response="[RESPONSE_PLACEHOLDER]", atoms=[ATOMS_PLACEHOLDER])
```

### A.7 V3 — PROSE AUDIT

**Input:** the response. **Output:** three 1–5 scores and three span lists, every span quoted verbatim. **Decision:** majority per facet. 2,455 characters. Placeholder: `[RESPONSE_PLACEHOLDER]`. The list of spans that must *not* be flagged as leakage is the calibration described in §6.

```
Let's define a function named check_response(response: str).
The function audits a generated response and returns a JSON object:
  {"fluency": 1-5, "formality": 1-5, "organization": 1-5,
   "leakage": [list of quoted spans], "hedging": [list of quoted spans],
   "artifacts": [list of quoted spans]}

- fluency, formality, organization: 5 = indistinguishable from expert human
  prose; 1 = badly degraded.
- leakage: any span revealing the response was generated from an annotation
  plan. Read this NARROWLY: naming a relation type or sense; naming the plan,
  the strength bands, or the validity labels; referring to "claims", "atoms" or
  "statements" as the objects the text was assembled from; or any label, tag,
  numeric index, or bracketed marker. Do NOT flag a connective or coordination
  that merely realizes a relation - "either X or Y", "at least one of X or Y",
  "at least one of these was true", "perhaps both", "one or both",
  "the two cannot both be true, and one of them must hold",
  "one of these accounts must be wrong", "although X, Y",
  "before", "after", "subsequently", "earlier than", "which postdates",
  "as indicated by", "as shown by", "which indicates", or an
  explicit statement that an ordering carries no causal or inferential force. A
  writer is REQUIRED to use those, and a coupling over two claims' joint truth
  cannot be expressed without one. Every span you report must be QUOTED VERBATIM
  from the response; if you cannot copy it out of the text, it is not there and
  must not be reported. Do NOT
  flag ordinary scholarly vocabulary used about the subject matter, such as
  "claim", "outcome", "reading", "account" or "finding".
- hedging: a span using one of these EXACT words: "assume", "might",
  "possibly", "allegedly", "supposedly", "reportedly", "it is claimed". The list
  is CLOSED - quote the word you found, and flag nothing that does not contain
  one of them. Uncertainty a writer is REQUIRED to express is not hedging: "at
  least one of X or Y, perhaps both" and "either X or Y" state the logical
  strength of a relation, not doubt about the content, so do NOT flag them.
- artifacts: enumerated list structure, headings, repeated template phrasing, or
  a sentence that reads as a stitched-together claim rather than prose.

Return the JSON object only. Note that your response will be passed to the
python interpreter, SO NO OTHER WORDS!

check_response(response="[RESPONSE_PLACEHOLDER]")
```

### A.8 V4 — CLAIM COVERAGE

**Input:** the response and its claims. **Output:** a status and quoted span per claim. **Decision:** majority per claim over the gating statuses. 1,403 characters. Placeholders: `[RESPONSE_PLACEHOLDER]`, `[ATOMS_PLACEHOLDER]`.

```
Let's define a function named check_atom_coverage(response: str,
atoms: list[str]).
For each atom, the function decides whether the response asserts it, returning a
JSON list of objects:
  {"index": i,
   "status": one of ["asserted", "altered", "missing", "merged"],
   "span": the quoted span of the response that asserts it, or null,
   "position": the 1-based index of that span among the response's sentences}

- "asserted": the response asserts the atom's full content, in any wording.
- "altered": the response asserts something related but changes the atom's
```

```
  content (a different number, date, entity, or polarity).
- "missing": the response does not assert the atom.
- "merged": the atom's content has been absorbed into another atom's, so it is no
  longer separately assertable - you could not negate this atom alone and leave
  the other standing. Reserve this for genuine loss of content.
  Two atoms joined by a connective are NOT merged: "either X or Y", "at least one
  of X or Y", "although X, Y", "X, whereas Y", and "X; that is, Y" each assert
  BOTH atoms, because each remains independently true or false. Mark both
  "asserted", with the span covering the part that asserts each one.

Return the JSON list only. Note that your response will be passed to the python
interpreter, SO NO OTHER WORDS!

check_atom_coverage(response="[RESPONSE_PLACEHOLDER]", atoms=[ATOMS_PLACEHOLDER])
```

### A.9 Retry feedback

When an attempt fails, the prompt is re-issued with one of two notes appended, the failure reason substituted for `{reason}`. The *format* note follows a parse failure; the *content* note follows a gate failure and tells the model its format was correct. Neither note is a separate prompt: each is appended to the stage's own prompt, and a test asserts that neither contains text that could route the retry to another prompt's parser.

**Format note (parse failure).**

```
NOTE: your previous attempt could not be read by the automated reader.
Reason: {reason}
Emit only the output described above, in exactly the required format.
```

**Content note (gate failure).**

```
NOTE: your previous attempt was well-formed, but its content did not meet a quality
↪  requirement.
Problem: {reason}
Keep the same output format and revise the content so this is fixed.
```

## B Relation-mining and baseline prompts, verbatim

These prompts are not part of the benchmark: they are the prompts of the measures we evaluate on it in §10, reproduced so that those results can be replicated. Placeholders here use `{{name}}` (mining) or `{name}` (judges) and are filled per call.

### B.1 Mining prompt A — sense and coupling

Issued once per candidate claim pair, with the full response as context, so that the model asserts only relations the response draws. The bracketed `[coupling=...]` tag is read from the output, and its token probabilities give the type confidence. The released mined arms use this variant with the full sense menu.

```
Instructions:
You are given the full RESPONSE a model produced, and two atomic claims, A and B, both taken
↪  FROM THAT RESPONSE, in their order of appearance (A comes before B). Your task is to
↪  decide the discourse/logical relation FROM A TO B, following the steps below.

IMPORTANT -- ground your decision in the response. Assert a coupling ONLY if the response
↪  ITSELF draws that connection between A and B (as written, or as a clear step in the
↪  author's argument/narrative). Do NOT assert a relation that is merely plausible in
↪  general but that the response does not actually make. If A and B both appear in the
↪  response yet the response draws no logical or discourse dependence between them, the
↪  answer is None.

1. Reason step by step, referring to the response: does the response present A as causing,
↪  enabling, providing evidence for, restating, elaborating, temporally preceding,
↪  contrasting with, or contradicting B? Is B a claim the response later withdraws or that a
↪  holding resolves? Consider the direction (A to B). If the response links A and B only
↪  indirectly through other claims, or not at all, that is None.

```

```
2. Name the DISCOURSE SENSE, one of: Cause-Effect, Effect-Cause, Evidence, Condition, Restatement, Instantiation, Contrast, Concession, Alternative, Disjunction, Precedence, Succession, None.
   - Cause-Effect: A causes/leads to B. Effect-Cause: A is the effect, B its cause.
   - Evidence: A provides evidence for B. Condition: A is a condition for B.
   - Restatement: A and B assert the same thing. Instantiation: A is a general claim, B a specific instance (or vice versa).
   - Contrast: A and B are in opposition but NOT exhaustive (they need not cover all possibilities; both could conceivably be false).
   - Concession: A and B are in tension but the text concedes/resolves it ("although A, still B", or a holding settles it).
   - Alternative: A and B are EXHAUSTIVE competing options -- EXACTLY ONE holds (they are mutually exclusive AND together cover the possibilities: not both, and not neither). E.g. "no one was harmed" vs "three people died"; "the cause was pilot error" vs "the cause was a metallurgical defect".
   - Disjunction: AT LEAST ONE of A and B holds (they may both hold, but the response rules out neither being true) -- e.g. two supporting findings at least one of which must be present.
   - Precedence/Succession: A and B are ordered in time with no truth dependence.
   - None: the response draws no logical or discourse dependence between A and B.

3. Map the sense to a COUPLING, one of: entailment, contradiction, equivalence, exclusive, co_necessity, none.
   - Cause-Effect, Effect-Cause, Evidence, Condition, Instantiation -> entailment
   - Restatement -> equivalence
   - Contrast, Concession -> contradiction
   - Alternative -> exclusive       (exactly one of A, B is true)
   - Disjunction -> co_necessity     (at least one of A, B is true)
   - Precedence, Succession, None -> none
   Prefer "exclusive" over "contradiction" when the two claims are not just incompatible but EXHAUSTIVE (one of them must be true); prefer "contradiction" when they merely cannot both hold but could both be false.

4. Give your final answer as two bracketed tags on ONE line, sense first:
[sense=Cause-Effect] [coupling=entailment]
A JSON object {"sense":"Cause-Effect","coupling":"entailment"} is also acceptable.

Use the following examples to better understand your task.

Example 1 (the response makes the causal link):
RESPONSE: The company launched a flawed product last quarter. Reviewers panned it, returns spiked, and the company's stock price fell 15 percent over the same period.
A: The company launched a flawed product last quarter.
B: The company's stock price fell 15 percent last quarter.
1. Reasoning: the response presents the flawed launch as the head of a chain (panning, returns) that ends in the stock decline, so the response itself draws a causal link from A to B.
2. Discourse sense: Cause-Effect.
3. Coupling: A causing B is a positive inferential link, i.e. entailment.
4. Final answer:
[sense=Cause-Effect] [coupling=entailment]

Example 2 (both claims present, but the response draws NO connection -> None):
RESPONSE: The quarterly report was published in April. Separately, the annual audit was scheduled for December. The two processes are run by different teams and were not related this year.
A: The quarterly report was published in April.
B: The annual audit was scheduled for December.
1. Reasoning: both claims appear in the response, and one might imagine a reporting-to-audit link in general, but this response explicitly treats them as separate and unrelated. The response draws no dependence from A to B.
2. Discourse sense: None.
3. Coupling: no dependence the response asserts, i.e. none.
4. Final answer:
[sense=None] [coupling=none]

Example 3 (the response states an EXHAUSTIVE alternative -> exclusive):
RESPONSE: The official statement said no one was harmed in the incident. However, the coroner's report confirmed that three people died in the incident.
A: No one was harmed in the incident.
B: Three people died in the incident.
1. Reasoning: the response sets A and B against each other ("However, ...") and they cannot both be true; but they also cannot both be false -- either people were harmed or they were not -- so exactly one holds. This is exhaustive, not a mere contrast. No holding resolves it.
2. Discourse sense: Alternative.
3. Coupling: exactly one of A, B is true, i.e. exclusive.
4. Final answer:
[sense=Alternative] [coupling=exclusive]

Example 4 (at least one must hold -> co_necessity):
RESPONSE: The defect was caught in review: at least one of the two independent checks -- the vibration analysis or the metallurgical assay -- flagged it.
```

```
A: The vibration analysis flagged the defect.
B: The metallurgical assay flagged the defect.
1. Reasoning: the response asserts the defect WAS caught by at least one check, so A and B cannot both be false; but both could hold (both checks may have flagged it). This is a disjunction, not an exclusion.
2. Discourse sense: Disjunction.
3. Coupling: at least one of A, B is true, i.e. co_necessity.
4. Final answer:
[sense=Disjunction] [coupling=co_necessity]

Your task:
RESPONSE: {{response}}
A: {{atom_a}}
B: {{atom_b}}
```

### B.2 MINING PROMPT B — CONDITIONAL STRENGTH

Issued for each pair that prompt A coupled. The strength is the renormalized probability of the answer token, $P(\text{Yes})/(P(\text{Yes}) + P(\text{No}))$.

```
Instructions:
You are given the full RESPONSE, two claims A and B drawn from it, and a {{coupling}} relation that holds from A to B. Assuming A is TRUE, judge -- IN THE CONTEXT OF THIS RESPONSE -- whether the relation's implication about B is credible: at least plausible / more likely than not, NOT whether it is certain.

- entailment or equivalence: given A and how the response uses it, is B at least plausibly TRUE (more likely than not)?
- contradiction or exclusive: given A and how the response uses it, is B at least plausibly FALSE (more likely than not)? (For "exclusive", A and B are exhaustive alternatives, so A being true makes B false.)
- co_necessity: A and B are a pair of which at least one holds. Given the response rules out "neither", is it at least plausible that B holds when A does NOT?

Answer with a SINGLE WORD, the very first word of your reply: Yes or No.
- Answer "Yes" if the implication is credible/plausible (even if not certain).
- Answer "No" only if the implication is implausible or the response does not actually support it.
Do not output anything before the word Yes or No.

Example 1 (response supports a near-certain entailment):
RESPONSE: The new alloy is chemically identical to the certified reference alloy, so it meets the certified reference specification.
A: The new alloy is chemically identical to the certified reference alloy.
B: The new alloy meets the certified reference specification.
coupling: entailment
Answer: Yes

Example 2 (weak but plausible, and the response draws the link -- still Yes):
RESPONSE: The company launched a flawed product last quarter, and its stock price fell 15 percent over the same period.
A: The company launched a flawed product last quarter.
B: The company's stock price fell 15 percent last quarter.
coupling: entailment
Answer: Yes

Example 3 (clear contradiction stated by the response):
RESPONSE: The official statement said no one was harmed, but three people died in the incident.
A: No one was harmed in the incident.
B: Three people died in the incident.
coupling: contradiction
Answer: Yes

Your task:
RESPONSE: {{response}}
A: {{atom_a}}
B: {{atom_b}}
coupling: {{coupling}}
Answer:
```

### B.3 G-EVAL COHERENCE JUDGE

The SummEval coherence rubric as used by G-Eval (Liu et al., 2023; Fabbri et al., 2021), adapted from summaries to responses, rated 1–5 and normalized.

```
You will be given one response written for a query.

Your task is to rate the response on one metric.

Evaluation Criteria:

Coherence (1-5) - the collective quality of all sentences. We align this dimension
with the DUC quality question of structure and coherence, whereby the response
should be well-structured and well-organized. The response should not just be a
heap of related information, but should build from sentence to sentence to a
coherent body of information about a topic.

Evaluation Steps:

1. Read the response carefully and identify the main topic and key points.
2. Check whether the response presents them in a clear and logical order, and
   whether later statements are consistent with earlier ones.
3. Assign a score for coherence on a scale of 1 to 5, where 1 is the lowest and 5
   is the highest.

Response:

{response}

Reason briefly, then give your rating as a single digit in square brackets, for
example [3].
```

### B.4 DIRECT COHERENCE JUDGE

The no-rubric control.

```
Read the following response and rate its logical coherence -- how well it hangs
together as an argument, and whether it contradicts itself -- on a scale of 1 to 5,
where 1 is the lowest and 5 is the highest.

Response:

{response}

Give your rating as a single digit in square brackets, for example [3].
```

## C THE PROMPT SPECIFICATION AND SUBSTITUTION CONTRACT

Each pipeline prompt's numbered-instruction count and required placeholder set are declared in a specification table, and a test asserts the declaration against the prompt strings.

| Prompt | Characters | Instructions | Required placeholders |
|---|---|---|---|
| P1 | 1,902 | 9 | `TOPIC` |
| P2 | 3,752 | 7 | `QUESTION` |
| P3 | 7,778 | 10 | `CLAIMS`, `QUESTION` |
| P4 | 5,571 | 8 | `PLAN`, `QUESTION` |
| P5 | 2,592 | 5 | `OPERATOR`, `PLAN`, `RESPONSE` |
| V1 | 4,086 | 0 | `ATOMS`, `RESPONSE` |
| V3 | 2,455 | 0 | `RESPONSE` |
| V4 | 1,403 | 0 | `ATOMS`, `RESPONSE` |

Instructions are counted by matching lines that begin at column one with a digit, a period and a space, so indented sub-items do not inflate the count. Placeholders are extracted by matching `[A-Z_]+_PLACEHOLDER` in brackets. Substitution enforces a *bidirectional* exact match between declared placeholders and supplied values: a missing name and an extra name are both errors, since a typo in either direction would otherwise ship a prompt containing a literal `[FOO_PLACEHOLDER]`, which a model will treat as content.

**A validator that was specified and removed.** The original design included a fourth validator to adjudicate exhaustiveness. It is deliberately absent: exhaustiveness is derived from the sense by the compile map, so the validator would have re-derived from prose what the scheme fixes by construction,

and could have contradicted it. Exhaustiveness is instead checked by V1's sense-recovery rate, and by the validation study's both-false and both-true questions (§11.2).

# D HUMAN-STUDY MATERIALS

## D.1 PILOT ANNOTATOR BRIEF

The pilot's full written brief, given to every annotator before the first screen. [Validation study: add the core-session and instruction-test briefs, and a screenshot of the claim-pair and highlighted-change screens, once finalized.]

> **The task.** Each screen shows two responses, A and B, answering the same question. You decide which one is *more logically coherent*, or whether they are about equal.
>
> **What "logically coherent" means here.** A response is logically coherent when the things it says can all be true together, and the connections it draws between them hold up. Ask yourself: if I believed every claim in this response, would I be believing something consistent? Look for contradictions; connections that do not follow; tensions the text raises and then leaves hanging — versus tensions it raises and then settles, which is fine; and broken chains of reasoning.
>
> **What is not the target.** Do not reward length; fluency, style or polish; vocabulary or sophistication; or topic knowledge — a response can be perfectly coherent and factually wrong. Sentence order in itself does not make a response less coherent, unless the new order actually breaks a connection. That last point is a real question we are trying to answer, so please judge it honestly.
>
> **The three answers.** A; B; or about equal — you cannot tell them apart on coherence, or any difference is about style, length or wording. "About equal" is a real answer, not a cop-out; but if you see a difference in consistency, however small, pick a side and mark low confidence.
>
> **Confidence and reason.** Rate confidence low, medium or high, and give one sentence saying what drove the choice, ideally naming the specific claims or connection.

## D.2 PILOT SCREEN SELECTION

| Family | Ladder | Topic | Pairs |
|---|---|---|---|
| f014, f013, f012 | CONFLICT | Education, Economics, Cybersecurity | $(\rho_0, \rho_1)$, $(\rho_0, \rho_4)$ |
| f009, f005 | CHAIN | Civil Engineering, Astronomy | $(\rho_0, \rho_1)$, $(\rho_0, \rho_4)$ |
| f007 | ORDER | Botany | $(\rho_0, \rho_1)$, $(\rho_0, \rho_4)$ |
| f003 | CONTROL | Art | $(\rho_0, \rho_2)$, $(\rho_1, \rho_2)$ |

Families were chosen where the gold-graph measure and the strongest evidence-free baseline disagree most, including one (f012) on which the baseline wins. The CONTROL family's endpoint pair is byte-identical by design, so its two screens compare each meaning-preserving edit with the untouched base.